

BUY

INITIATION OF COVERAGE

Financial Summary

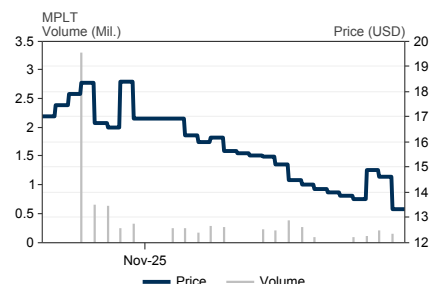
Changes	Previous	Current
Rating		Buy
Target Price	—	\$28.00
FY25E Revenue	—	\$0.0
FY26E Revenue	—	\$0.0
FY25E EPS	—	\$(5.34)
FY26E EPS	—	\$(2.61)

Price (11/20/25):	\$13.29
52-Week Range:	\$21 - \$13
Market Cap.(mm):	742.9
Shr.O/S-Diluted (mm):	55.9
Avg Daily Vol (3 Mo):	NA
FYE	Dec

52-week range since shares began trading 10/27/25

	2024A	2025E	2026E
Revenue	0.0	0.0	0.0
EPS	\$(6.27)A	\$(5.34)	\$(2.61)

Price Performance



Initiating Coverage With a Buy Rating: Following the Cobenfy Playbook with a Potentially Differentiated Next-Gen Muscarinic

Summary

We are initiating on MPLT with a Buy rating and a \$28 Target Price. Bottom line: we believe MPLT represents an attractive opportunity as a potential second-mover in the muscarinic space, where despite the slower than expected launch of Cobenfy, we still see a very large revenue opportunity. Specifically, we think MPLT's ML-007C-MA leverages learnings from KRTX/CERE/NBIX (i.e. M1/M4 agonism may be the right mechanism), but with potential avenues for differentiation, specifically around more convenient dosing and improved tolerability. For the stock, the next catalyst is the BMY ADEPT-2 readout, an important event for the space where Street expectations are fairly low--thus the risk/reward for MPLT is upside biased, and even if the study were negative, full data will be key to truly understand the broader implications. Meanwhile, MPLT is running two ph2 trials of ML-007C-MA in schizophrenia and ADP, with key data in 2H26 and 2H27, respectively.

Key Points

MPLT's ML-007C-MA leverages the learnings of KRTX, CERE and NBIX, but with room for potential differentiation. Since the early successes of Karuna and Cerevel with their muscarinic drugs KarXT (M1/ M4 agonist w/ peripheral antagonist) and emraclidine (M4 PAM), there's been the longstanding debate on what constitutes the ideal muscarinic pharmacology to maximize efficacy while minimizing side effects. Initially it seemed like M4-selectivity may be the best approach, but since we've seen late stage failures for emraclidine, along with mixed data from NBIX's NBI-568 (M4 agonist), which all together suggest that having M1 activity is helpful and maybe even essential. To this point, MPLT's ML-007C-MA is highly potent at both M1 and M4 receptors, and from a tolerability perspective seemingly has a better PK match with its complementary peripheral antagonist (fesoteridine) which may help mitigate side-effects. And while this needs to play out in the ph2 data, PK/PD modeling suggests that ML-007C-MA may (unlike Cobenfy) be efficacious at a QD dose, which, along with its lack of food effects, are according to KOLs key differentiators in schizophrenia where compliance is an issue.

Even with the slower than expected launch of Cobenfy in schizophrenia, we still think the revenue opportunity here is significant. So far, Cobenfy uptake in schizophrenia has tracked better than prior launches of atypicals, but is still well below what we had modeled when we covered KRTX (before the BMY takeout). The question here is whether we should re-rate our expectations for peak sales. In our view, unlike mood disorders where demand can be very patient driven, there's reason to think that drug launches in schizophrenia may naturally be slower, with the prescriber base being somewhat more cautious around switching (or what to use during an acute relapse). Our survey work and recent KOL checks still suggest that muscarinics will become a staple of the standard-of-care over time, and with meaningful penetration in the treatment-refractory population, there's the potential for a blockbuster opportunity.

BMY's ADEPT-2 readout is the next catalyst for the space; expectations are low and if anything this is more of an overhang for MPLT, making the risk/reward highly upside biased. In our view, we believe muscarinics should work in ADP as long as high enough (i.e. close to schizophrenia-level) exposures are well tolerated. This is the central question to Cobenfy's probability-of-success, as even with TID dosing in the elderly (vs BID in schizo), side effect rates look materially higher. BMY's messaging around this readout hasn't been all that confidence inspiring, but if anything this has become an overhang for MPLT to the point that (1) surprisingly positive data should send shares materially higher while (2) a negative readout would need to be interpreted with nuance, though we may need to wait to get full data if the trial fails.

We derive a \$28 TP. This reflects a 50% PoS in schizophrenia modeling ~\$1.2B in 2038E gross sales, along with a 15% PoS for ADP where we model a ~\$1.9B peak.

MPLT is funded through their two key readouts in 2H26 and 2H27, while they also have another program in autism spectrum disorder that largely represents upside to our model.

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All relevant disclosures and certifications appear on pages 66 - 68 of this report.

Investment Thesis

We rate MPLT shares Buy. MapLight Therapeutics is a clinical-stage biotechnology company focused on developing drugs for neuropsychiatric disorders. Our bullish thesis on MPLT is predicated on the potential of ML-007C-MA, a co-formulated drug consisting of the M1/M4 muscarinic agonist ML-007 and peripheral anticholinergic fesoterodine. MPLT's ongoing ph2 in schizophrenia is de-risked by KarXT's prior success leveraging a similar mechanism, though MPLT is also pursuing a ph2 in Alzheimer's disease psychosis where KarXT clinical data are still upcoming. Despite MPLT entering several years behind KarXT in an already crowded schizophrenia market, the "bar" for success is low and we've seen blockbuster sales for prior antipsychotics with minimal differentiation. In our view, less frequent dosing may be sufficient in itself for prescribers to select ML-007C-MA over KarXT, especially in ADP where KarXT's thrice daily dosing can be cumbersome. Ultimately though, this is CNS and good drugs with strong scientific rationale can still fail in clinical trials, but we think the risk/reward is upside-biased for this stock.

Target Price Methodology/Risks

We derive a \$28 target price for MPLT shares based on our DCF analysis. Our DCF uses a 11% discount rate and 0% terminal growth rate, aligned with other similar stage companies in our coverage universe. In our model we risk-adjust MPLT revenues in SCZ and ADP at 50% and 20%, respectively, which reflects the level of scientific rationale and de-risking clinical data generated to date. We view both the SCZ and ADP as significant commercial opportunities if MPLT is successful. Our terminal growth rate of 0% takes into consideration the longer DCF forecast period extending up to 2038 which may be close to capturing peak sales of ML-007C-MA.

There are many risks to an investment in MapLight Therapeutics, including clinical, regulatory, and commercial risks, as well as the risk of a dilutive equity offering. Specifically, MPLT investors face the risk that ML-007C-MA and other pipeline programs could fail in clinical trials, be impacted by a safety setback, and/ or be unable to achieve FDA approval. Beyond clinical trial and regulatory risks, MapLight, like other biotechnology companies, faces significant commercial risks due to competition, pressure from insurers, and/or a variety of other factors. Any of the above variables could impact the stock and our target price significantly.

Company Description

MapLight Therapeutics, Inc is a clinical-stage biopharmaceutical company focused on advancing multiple drug candidates for neuropsychiatric disorders. Their lead candidate, ML-007C-MA, is a combination of a CNS-penetrant procholinergic compound with a peripherally-restricted anticholinergic compound to mitigate side effects, similar to KarXT. ML-007C-MA is currently being evaluated for the treatment of both schizophrenia and Alzheimer's disease psychosis in separate ph2 studies. MapLight also has other drugs in the pipeline including ML-004 for sociability/irritability associated with Autism Spectrum Disorder.

Initiating Coverage With A Buy Rating On MapLight Therapeutics, Inc

November 21, 2025

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All prices reflect the close of 11/20/2025, unless otherwise indicated.

MapLight Is a Clinical-Stage Biotech Developing an M1/M4 Agonist with a Peripheral Antagonist, In Schizophrenia and Alzheimer's Disease Psychosis

MapLight Therapeutics is a clinical-stage biopharmaceutical company developing their lead candidate ML-007C-MA, consisting of an M1/M4 agonist ML-007 co-formulated with a peripheral anticholinergic (PAC) fesoterodine, for schizophrenia and Alzheimer's disease psychosis. They're essentially following in the footsteps of KRTX (later acquired by BMJ), the company that pioneered the combination of pro- and anti-cholinergics with the development of Cobenfy (KarXT) that is now approved in schizophrenia and currently undergoing pivotal studies for ADP. Because MPLT leverages the same M1/M4 targets with a highly potent molecule, we think it's realistic that they'll see similar efficacy in schizophrenia, whereas ABBV/CERE and NBIX have (so far) come short. The thesis though is that MPLT's drug, relative to Cobenfy, can improve upon tolerability and dosing (frequency and food effects), which are important in schizophrenia but also may be key to patient convenience and arguably even success in Alzheimer's Disease Psychosis. Lastly, MPLT also has a ph2 program in ASD and several preclinical-stage assets, although those are not as de-risked and less core to our investment thesis.

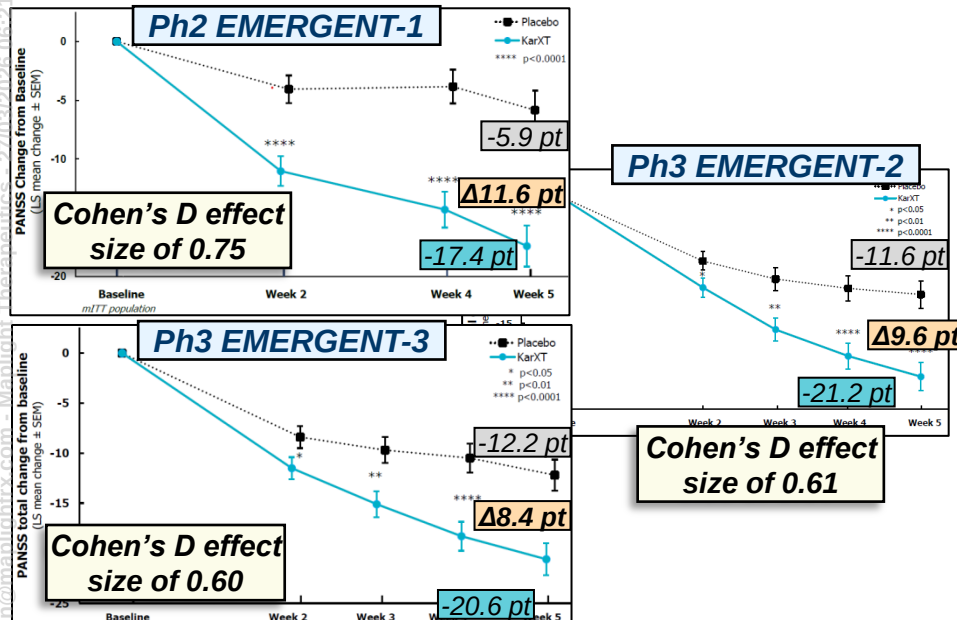
Program	Circuit	Indications	Preclinical	Phase 1	Phase 2	Phase 3	Anticipated Milestones
ML-007C-MA M/M ₄ agonist co-formulated with PAC	Direct and Indirect Pathways	Schizophrenia	ZEPHYR				Topline results in 2H 2026
		Alzheimer's Disease Psychosis	VISTA				Topline results in 2H 2027
ML-004 5-HT _{1B/1D} agonist	Dorsal Raphe to Nucleus Accumbens	Autism Spectrum Disorder Sociability/Irritability	IRIS				Topline results in 2H 2026
ML-021 M ₃ antagonist	Direct Pathway	Parkinson's Disease					IND-enabling studies complete in 2H 2026
ML-009 GPR52 PAM	Indirect Pathway	Hyperactivity/Impulsivity					Preclinical candidate nomination in 2026

Source: MapLight Materials; Stifel Research.

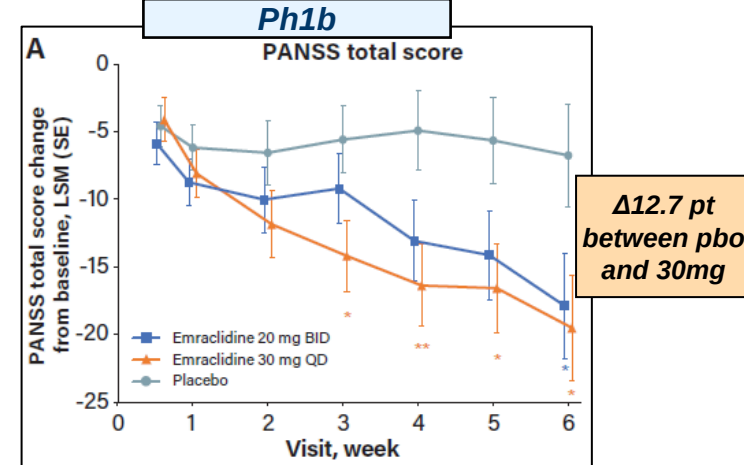
Goes Without Saying: KRTX and CERE Show the Upside Potential of a Successful Muscarinic

On the back of positive clinical data, Karuna and Cerevel were acquired for almost \$14B and almost \$9B, respectively. This was driven by the perceived commercial potential of a novel mechanism in schizophrenia, while there was definitely optionality ascribed to other indications including ADP. Again, the setback with emraclidine has opened the door for others to become a second mover after Cobenfy, and/or have a best-in-class muscarinic. As we think about these below valuation comps for a MPLT bull case, it is still a moving target as we see how the Cobenfy launch in schizophrenia goes, and whether the ADEPT-2 ADP trial works. But as it stands the current MPLT valuation is a small fraction of these deal values below, so we believe there's an attractive risk/reward here for the stock into multiple readouts over the coming years.

BMJ Acquired KRTX for \$14B



ABBV Acquired CERE for \$8.7B



CERE only had a small study with 81 patients 1:1:1 randomized to placebo, 30mg QD, and 20mg BID, yet garnered enough interest for a billion-dollar buyout

MPLT Leverages the Learnings from KRTX, CERE, and NBIX – By Design, ML-007C-MA Is On the Right Side of the Various “Ideal Muscarinic” Debates

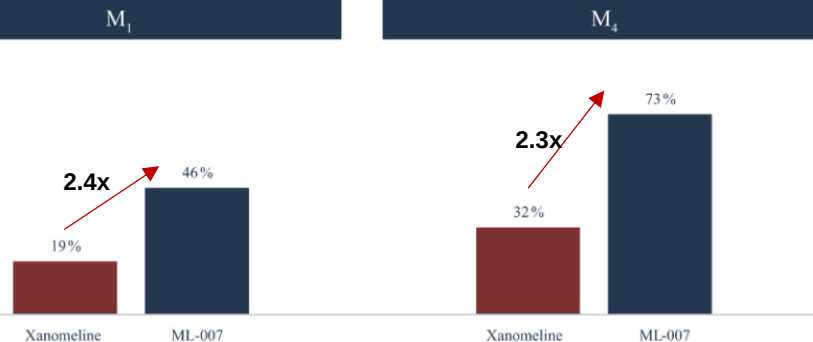
As the muscarinic space has become exciting, multiple other companies have tried to follow in KRTX’s footsteps but have encountered setbacks. For example, although the exact reasons for ABBV/CERE’s emraclidine’s failure in ph2 are unknown (probably multiple factors), we think it’s reassuring that MPLT’s drug is more similar to Cobenfy than emraclidine: (1) the importance of hitting M1 on top of M4 is unclear, but it’s less risky to just target both which is what ML-007 does, and (2) agonists like ML-007 can exert their effects independently, vs PAMs that need the natural muscarinic ligand. Meanwhile, MPLT is hoping to achieve more convenient dosing than Cobenfy which is BID in schizophrenia and TID in ADP. Based on ph1 PK modeling, they believe it’s possible that QD dosing will be sufficient, though MPLT is advancing both a QD and BID regimen in the SCZ ph2 (and BID in ADP) to be safe. Bottom line: none of these factors necessarily guarantee success, but we feel more confident extrapolating the Cobenfy data to MPLT’s molecule, than we would for say NBIX (selective M4 agonist) or ABBV/CERE (selective M4 PAM).

PAM 	<i>While the reason for emraclidine’s failure can be debated, from a biological perspective, PAMs require the natural ligand to be present near the receptor whereas agonists do not. So, agonists are more likely to be de-risked and efficacious under broader set of circumstances.</i>	Agonist 
M4 Selective 	<i>Perhaps a more widely debated topic is the importance of M1: some animal experiments show that M1 is necessary for antipsychotic efficacy while others suggest it’s incremental. But Cobenfy (M1/M4) is the only one that has succeeded in ph3 so far--pursuing a similar approach is less risky.</i>	M1/M4 Selective 
No Anticholinergic 	<i>Both KRTX and MPLT use a peripheral antagonist to minimize side effects, while the M4-only candidates do not. Emraclidine did show a better tolerability profile than Cobenfy, but they may have been underdosing and we haven’t seen details from the ph2.</i>	With Anticholinergic 
Twice Daily Dosing 	<i>Less frequent dosing is always better, especially in a neuropsychiatric disease where compliance is an issue. Pill burden also matters when a food effect needs to be navigated. Note that MPLT is aiming for once daily dosing in SCZ, though they are advancing both QD and BID dosing in ph2.</i>	Once Daily Dosing 

ML-007 Is A Brain-Penetrant M1/M4 Agonist That Is Seemingly More Active Than Xanomeline Preclinically

MPLT selected ML-007 based on favorable physical and chemical properties. In addition to solubility, high oral bioavailability, and lack of first-pass metabolism through liver, the compound is also more active: *in vitro* assays showed greater than two-fold peak activity at both M1 and M4 receptors compared to xanomeline, the pro-cholinergic component of *Cobenfy*. Meanwhile, from *in vivo* animal studies, MPLT extrapolated that ML-007 had ~10x greater effect vs xanomeline at similar doses. Given that xanomeline alone and xanomeline+trospium (*Cobenfy*) showed efficacy in schizophrenia, we can be at least cautiously optimistic that a drug like ML-007 that hits M1 and M4 at the same ratio but harder should also work, though a peripheral anticholinergic will be needed to mitigate AEs. The open question is whether greater potency will drive greater efficacy – at least in schizophrenia we'd be cautious here, though there may be more room to differentiate on the efficacy side in ADP as things like the dosing profile and the therapeutic index become more important.

In Vitro Assay Shows Superior Activity Over Xanomeline



1. Represents data normalized to responses of control agonist, oxotremorine, in human GTPγS M₁ and M₄ in vitro assays.

In Vivo Animal Models Point To Greater Activity Over Xanomeline



- Amphetamine-induced hyperlocomotion
- PCP-induced hyperlocomotion
- Conditioned avoidance response

Based on PD and CSF exposures of ML-007, it has ~10x greater effects at same dose compared to xanomeline

Preclinical evidence suggests that MPLT's ML-007 induces greater M1/M4 activity than *Cobenfy*'s xanomeline

ML-007 Will Be Complemented By Fesoterodine, A Well-Characterized Peripheral Muscarinic Antagonist

To counteract the peripheral cholinergic side effects that limited tolerability of xanomeline, Karuna coformulated the drug with peripheral antagonist trospium to dampen at least some of the side effects. Similarly, MPLT is opting to use fesoterodine, a different peripheral antagonist which they believe has favorable properties as well, including half-life and PK curves that match ML-007, with minimal DDIs. Fesoterodine itself has long been approved for overactive bladder and neurogenic detrusor overactivity and while there are associated adverse events, many of these are due to its anti-cholinergic effects that in the ph1 seem counterbalanced by ML-007. Bottom line: the idea here is to have a better PK match that has improved tolerability over KarXT – we think the data so far are supportive of this thesis, especially on specific AEs.

Fesoterodine Adverse Events

System organ class/Preferred term	Placebo N=554 %	Toviaz 4 mg/day N=554 %	Toviaz 8 mg/day N=566 %
Gastrointestinal disorders			
Dry mouth	7.0	18.8	34.6
Constipation	2.0	4.2	6.0
Dyspepsia	0.5	1.6	2.3
Nausea	1.3	0.7	1.9
Abdominal pain upper	0.5	1.1	0.5
Infections			
Urinary tract infection	3.1	3.2	4.2
Upper respiratory tract infection	2.2	2.5	1.8
Eye disorders			
Dry eyes	0	1.4	3.7
Renal and urinary disorders			
Dysuria	0.7	1.3	1.6
Urinary retention	0.2	1.1	1.4
Respiratory disorders			
Cough	0.5	1.6	0.9
Dry throat	0.4	0.9	2.3

Impact Of Muscarinics On Different Organ Systems

Organ System	Procholinergic	Anticholinergic
 mouth	Increased saliva production	Decreased saliva production
 stomach	Increased motility, more gastric acid	Decreased motility, less gastric acid
 eyes	Teariness	Dry eye, blurred vision
 bladder	Urinary frequency	Urinary retention

MPLT Has Completed Four Ph1 HV Studies To Extensively Optimize The Dosing Regimen For Ph2

MPLT conducted a fairly comprehensive ph1 program. In brief, they established that (1) co-administration of their procholinergic agent (ML-007) with peripheral anticholinergic (Fesoterodine) improved tolerability vs ML-007 alone, (2) ML-007-to-PAC plasma ratios between 100:1 and 600:1 were associated with fewer cholinergic AEs, (3) ML-007 and fesoterodine can be coformulated into a single bilayer tablet with favorable PK, (4) healthy adults were able to tolerate QD dosing with minimal titration, while elderly patients did best with BID dosing and ~1 week titration, and (5) absorption can be enhanced with food, so no fasting is required. Bottom line: the work so far has optimized the dosing going into schizophrenia, where QD and BID will be studied – whereas in ADP, BID may be optimal. Moreover, rates of certain AEs, especially anti-cholinergic, look better than Cobenfy with the caveat that the size of these cohorts in the ph1 is somewhat small.

	Study Design	Results And Implications
Study 001 (N=58)	- Single ascending doses of ML-007 oral solution in HV - Group 1 only received ML-007 between 0.8 to 49mg - Group 2 also received either placebo or PAC ER four hours prior to getting 32mg of ML-007	- ML-007 was generally well tolerated up to the 32mg dose - CSF concentration at this dose reached 50ng/mL - Administration of PAC improved tolerability
Study 011 (N=106)	- Different ratios of ML-007 to PAC were administered - Evaluated safety and tolerability of multiple doses in HV and elderly participants	- For single doses, up to 160 mg of ML-007 was well-tolerated if co-administered with PAC - For multiple doses, up to 111 mg of ML-007 was well-tolerated in HV/Elderly if co-administered with PAC
Study 012 (N=24)	- Open-label ph1 comparing ML-007 with and without PAC - Used extended release formulations	- AE rates lower with ER vs oral solution formulation - PK was measured to inform dosing strategy for next trials
Study 013 (N=82)	- Used the co-formulated bi-layer tablet formulation that is being advanced into the ph2 - Evaluated single and multi-dose regimens in HV and elderly participants - Food effects and titration schedules were tested	- ML-007 exposures were higher with meals - Tolerability was improved with single lower titration dose in adults and one week titration in elderly - Generally favorable tolerability with QD and BID dosing in adults and elderly patients, respectively

Altogether, MPLT's Drug Profile Should At Least Offer Incremental Improvements Over Cobenfy – And In Large Psych Markets, There's Room for Multiple Players

In an ideal scenario, there are multiple aspects where ML-007C-MA may improve on Cobenfy. That said, even incremental improvements can lead to meaningful market share and sales – i.e. using other atypical antipsychotics as an analog.

Easier to Achieve

**Ease of Use –
Beyond Mere
Convenience**

How Might ML-007C-MA Compare/Differentiate to Cobenfy

This is one of the more obvious benefits that ML-007C-MA has over Cobenfy given that MPLT has (1) the potential for QD dosing vs. BID in schizophrenia, (2) a single dose titration vs. 3-8 days, and (3) no food effects means MPLT can be dosed in fed state vs. required fasting. There's also the potential for BID dosing in ADP (vs TID Cobenfy), which with the lack of food effect and even slightly improved tolerability (below) may even support a higher probability-of-success in ADP. And while "convenience" may seem incremental, commercially, it's important in neuropsych where compliance can be an issue.

**Safety and
Tolerability**

Both KRTX and MPLT had the idea of counteracting the peripheral cholinergic AEs by combining a brain-penetrant pro-cholinergic with a peripheral anti-cholinergic. However, MPLT believes that based on their extensive ph1 testing, their ph2 doses should have the optimal ratio of pro-to-anti cholinergic activity that will further minimize side effects. In our view, the data so far support this thesis on at least some AEs, predominantly those that are anti-cholinergic.

**PANSS
Efficacy**

In vitro studies showed that ML-007 had ~2.5x greater intrinsic activity on M1 and M4 receptors than Cobenfy does, while in vivo animal studies showed ~10x greater activity at the same dose. We don't have receptor occupancy data for either drug though, so it's really hard to know whether greater in vitro activity will equate to superior efficacy. That said, greater activity is probably not a bad thing.

**Other
Subdomains**

MPLT has conducted animal studies that showed improvements on both spatial and social memory in a mouse model of Alzheimer's disease with ML-007 but not xanomeline, potentially due to greater M₁ activity for the former. Greater cognition benefits would be awesome, but this is a high bar and at least in schizophrenia, showing an independent cognitive effect is challenging.

Higher Risk/Harder to Say

Source: MapLight Materials; Stifel Research.

STIFEL

Taking A Deeper Look: MPLT Has Rigorously Optimized The Ratio Of ML-007 vs PAC

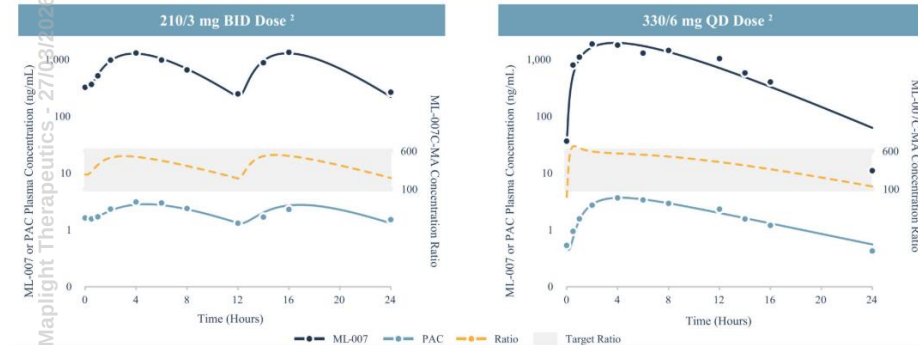
MPLT believes that ML-007:PAC plasma exposure ratios between 100:1 to 600:1 are the ideal range, determined empirically from their ph1 study (011) using measures of cholinergic activity (salivary volume changes) and frequency of adverse events, both procholinergic (nausea, vomiting, diarrhea, hyperhidrosis) and anticholinergic (dry mouth, tachycardia, dry eye). While this is a fair approach, the underlying assumptions for MPLT's drug to be more tolerable than Cobenfy are that (1) the ideal pro- to anti-cholinergic plasma ratios are translatable between HV and schizophrenia patients, and (2) Cobenfy does NOT fall within its ideal range. For context, Karuna (at PureTech originally) also did a ton of work on dose optimization, but they may have been constrained by the challenging med-chem profile of xanomeline, which is a great drug but also a very old drug. Our view is that MPLT has done good work here and that there is room to improve over KarXT/Cobenfy on AEs.

"Ideal" Ratio between Pro- and Anti-Cholinergics

Data Represents Steady State Plasma Concentrations From Study 013¹

210/3 mg BID Dose²

330/6 mg QD Dose²



Did KRTX Test Different Ratios For SCZ?

Study	Xanomeline Dose	Trospium Dose
Ph1 POC	75mg TID	-
	75mg TID	20mg BID
Ph1 Co-formulation	100mg BID	20mg BID
	125mg BID	30mg BID
	150mg BID	40mg BID
Ph2 and Ph3 studies	50mg to 125mg BID	20mg to 30mg BID

In the ph1 Cobenfy studies, only a few combinations of xanomeline and trospium doses were explored before the final 50mg/20mg, 100mg/20mg, and 125mg/30mg dose titration regimen was selected for ph2 and ph3 studies. Further optimization could be possible in theory.

From study 011, MPLT established the ideal plasma ratio between their two compounds, which their co-formulated tablet was able to conform to in study 013. Though a key question is whether this ratio of optimal tolerability will translate from HV to schizophrenia patients.

Ph1 Data Points To Better Tolerability For MPLT On At Least Some Aspects – Better on Anti-Cholinergic AEs Whereas Cholinergic Harder to Say (1/2)

To validate the claim that MPLT's drug has a more optimal ratio, we attempted to compare the actual safety profile of these drugs in the ph1 HV studies. KRTX has limited safety data from their ph1 HV studies: detailed cholinergic AEs were only listed for the 75mg TID/20mg BID dose which is slightly less exposure than the max approved dose (125mg/30mg BID). At a glance, MPLT's doses which were selected for ph2 showed more nausea but less vomiting, and similar rates of sweating. Though notably, constipation was another AE in Cobenfy's subsequent studies that was not observed with MPLT's drug so far; this could be a big deal since constipation was one AE for Cobenfy that didn't really "tolerate out". Altogether, this suggests MPLT may have a differentiated tolerability angle, but it is still early days.

Safety/Tolerability of ML-007C-MA in HVs

	Placebo (n=8)	210/3mg BID (n=6)	330/6mg QD (n=6)
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Most common TEAEs (>1 subj)

Nausea	0%	50%	50%
Dizziness	0%	50%	33%
Chills	0%	0%	33%
Hyperhidrosis	0%	17%	33%
Dyspepsia	0%	0%	33%

Safety/Tolerability of Cobenfy in HVs

	Xanomeline 75mg TID (n=33)	Xanomeline 75mg TID + trospium 20mg BID (n=35)
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Total cholinergic AEs

	64%	34%
Nausea	24%	17%
Vomiting	15%	6%
Diarrhea	21%	6%
Sweating	49%	20%
Salivation	36%	26%

* No constipation or vomiting events were reported at these doses

Early data suggests ML-007C-MA may have less vomiting and constipation than Cobenfy, but these data are early.

At the Same Time, AEs Confirm ML-007 Target Engagement with Muscarinic Receptors (2/2)

Without POC data in schizophrenia patients, there's always the question of whether ML-007 will be active in human brain since (1) behavioral signals in animals may not translate to humans, and (2) efficacy on PANSS cannot be assessed in ph1 HV studies. Nonetheless, we can glean insights from the safety data which shows adverse events (nausea, hyperhidrosis) and reported vital sign changes (heart rate, blood pressure increases) that are commonly associated with pro-cholinergic activity--even if these are mostly a consequence of peripheral muscarinic agonism, PK data from ph1 HV (study 013) shows meaningful CSF exposures for ML-007 that should also translate to central muscarinic activity. Blood pressure and HR changes are things that tolerate over time but are on target with the M4 receptor, per Karuna and Cerevel data.

Safety/Tolerability of ML-007C-MA in HVs

	Placebo (n=8)	210/3mg BID (n=6)	330/6mg QD (n=6)
Most common TEAEs (>1 subj)			
Nausea	0%	50%	50%
Dizziness	0%	50%	33%
Chills	0%	0%	33%
Hyperhidrosis	0%	17%	33%
Dyspepsia	0%	0%	33%

Safety/Tolerability of ML-007C-MA in Elderly

	Placebo (n=7)	210/3mg BID, 2-7d titration (n=11)
Most common TEAEs (>1 subj)		
Nausea	0%	27%
Dizziness	14%	27%
Headache	0%	36%
Hyperhidrosis	0%	27%
Tremor	0%	27%

* No constipation or vomiting events were reported at these doses

* One episode of vomiting and one episode of constipation at this dose

Increases in heart rate and blood pressure were also observed in HV and/or elderly patients--both are associated with pro-cholinergic activity.

Based on Their Preclinical Data, ML-007 Could Achieve Higher Activity Over Cobenfy On M1/M4 In The Brain

Meanwhile on efficacy, MPLT believes ML-007 is a more active cholinergic at both M1 and M4 receptors in the brain. Their in vitro data demonstrate that while ML-007 has a lower pEC50 (more drug needed to elicit 50% of peak activity), the actual peak activity at high enough concentrations is more than two-fold that of xanomeline. Furthermore, in vivo experiments validate MPLT's PK advantage: ML-007 achieves higher plasma AUC than Cobenfy regardless of IV or IP administration, plus higher CSF concentrations as well. 10-fold lower doses of ML-007 were able to achieve comparable behavioral changes in three different mouse models as well. Though translation of mouse-to-human potency has its caveats, we note that MPLT is advancing both 210/3mg BID and 330/6mg QD doses into ph2 vs Cobenfy's max 125/30mg BID dose.

Pharmacokinetics of ML-007 and Xanomeline in C57BL/6 Mice

Compound	Dose (mg/kg)	Route	Plasma AUC (h*ng/ml)	t _{1/2} (h)	CSF C _{max} (ng/ml)
ML-007	1	IV	442.4	1.01	-
	0.3	IP	229.3	-	43.2
Xanomeline	1	IV	47.9	0.35	-
	3	IP	28	-	ND

Table 1. Pharmacokinetic studies were performed at Sai Life Sciences. All samples were analyzed with the fit-for-purpose LC-MS/MS method (xanomeline LLOQ = 0.49 ng/mL for plasma and 0.99 ng/mL for CSF; ML-007 LLOQ = 5.05 ng/mL for plasma and CSF) with parameters estimated using Phoenix® WinNonlin (software Ver 8.3).

Whether administered IV or IP, ML-007 reaches higher plasma AUC levels than xanomeline. Even at 3mg/kg IP doses, no xanomeline was detected in CSF, but ML-007 was detected at 10-fold lower doses.

Xanomeline binds more potently at lower conc. than ML-007 in vitro, but the max activity it can elicit is lower than ML-007. With greater CSF exposure, ML-007 could drive greater activity on M1/M4 than xanomeline.

ML-007 Has Higher Intrinsic Activity Than Xanomeline

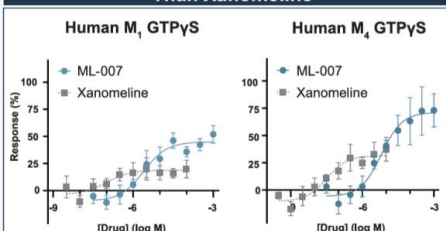
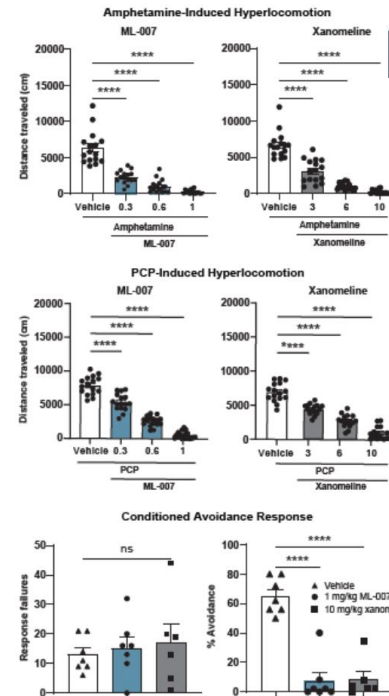


Figure 1. Data were normalized to the responses of control agonist, oxotremorine: 1 mM @ hM₁ and 100 μM @ hM₄.



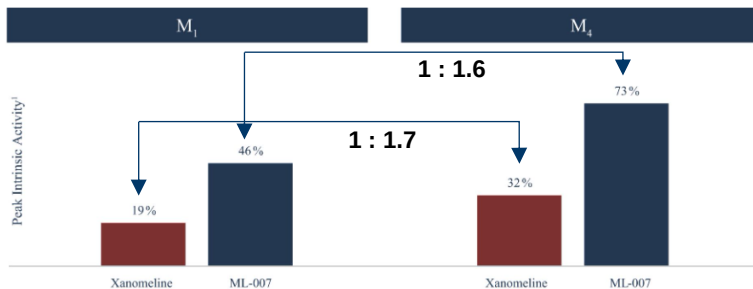
ML-007 Has Greater In Vivo Potency Than Xanomeline Across a Range of Classical Preclinical Models of Psychosis

In mice, MPLT was able to replicate the effects of xanomeline on three models of antipsychotic activity--amphetamine-induced and PCP-induced hyperlocomotion and conditioned avoidance response--with 10-fold lower doses of ML-007. Overall, the consistency of their results across models is reassuring and does validate their thesis on having a highly-active molecule.

Could Greater Activity On M1/M4 Translate To Greater Efficacy Than Cobenfy? May Be Tough in Schizophrenia, but in ADP There's Possibly More Oppty

After emraclidine's failure in ph2, the field continues to debate the potential of M4-only approaches in schizophrenia. Fortunately, that seemingly does not affect the probability of success for MPLT since ML-007 targets both M1 and M4--in fact, the similar ratio of M1:M4 activity between ML-007 and Cobenfy further supports the prospects of efficacy in schizophrenia. Moreover, preclinical studies suggest that at similar doses, ML-007 is up to 10x more active than Cobenfy, though the key question is whether we've hit a ceiling on PANSS reduction in human studies. Cobenfy already presents a high "bar" on PANSS improvement with a fairly robust effect size, but the ph2 and ph3 studies for Cobenfy and xanomeline only evaluated a single dose regimen and we don't have RO data, so we don't know if we're at the top of the dose-response curve. For MPLT, showing better efficacy is a source of optionality, albeit one that's hard to really diligence.

Ratio Of M1:M4 Engagement Remains Similar



Similarity in M1:M4 engagement to Cobenfy de-risks ML-007's efficacy in schizophrenia. Both also have off-target binding to M2, M3, and M5 receptors, though much less than M1/M4. Mgmt describes the ratio of off-target binding as similar between ML-007 and xanomeline.

Is There A Ceiling To Efficacy?

Study	Pbo-adj Difference from BL on PANSS Total Score	P-value	Cohen's d
Ph2 EMERGENT-1	11.6 pts	P<0.0001	0.75
Ph3 EMERGENT-2	9.6 pts	P<0.0001	0.61
Ph3 EMERGENT-3	8.4 pts	P<0.001	0.60

Recall that NBIX's ph2 for their M4 agonist showed an inverse dose response, but it's hard to know if this is biologically real, or just noise.

Meanwhile, ph2 and ph3 studies for xanomeline and KarXT only evaluated a single dose regimen, so it's hard to say if additional potency could lead to better PANSS. Ultimately, MPLT doesn't need to be more efficacious than KarXT to be commercially successful, in our view, but we would note that the KarXT data are quite good and already show strong efficacy relative to other antipsychotics.

And While Fesoterodine Crosses the BBB, MPLT Presents Their Case that This Should Have Minimal Impact on Efficacy

While greater pro-cholinergic activity in the brain is a positive for ML-007, we also need to be wary of potential anti-cholinergic activity of fesoterodine in the brain which could blunt the efficacy of the combo drug. Preclinical data suggests that fesoterodine (or specifically, its metabolite 5-HMT) could be up to 10x more CNS penetrant than trospium in animal models. Here, MPLT has done its own work to confirm fesoterodine won't impact ML-007 activity in the brain: in the amphetamine-induced hyperlocomotion mouse model, they tested increasing amounts of feso vs ML-007 and only started to observe a reduction in ML-007's effects at ML-007:PAC plasma ratios around 10:1 to 20:1--this gives us confidence that the 100:1 to 600:1 ratios being pursued in human studies should be in a safe zone.

ML-007 Has Greater CNS Penetration Than Feso

Drug (dose)	Plasma conc. (ng/mL)	CSF conc. (ng/mL)	CSF:Plasma
Trospium (1mg/kg; rat)	106 ± 85	0.33 ± 0.34	1:320
5-HMT* (0.3mg/kg; rat)	27.3 ± 1.3	0.74 ± 0.007	1:37
ML-007 (330mg; human)	~1200	~120**	~1:10

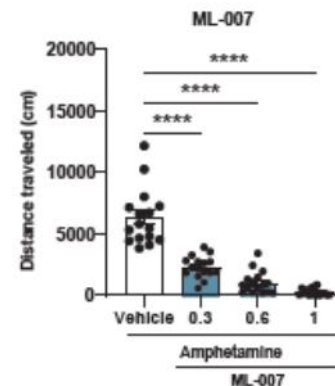
* 5-HMT is the active metabolite of fesoterodine

** Based on MPLT's PK models

A key assumption underlying MPLT's drug is that while pro- and anti-cholinergic effects are balanced in the periphery, ML-007 is more brain-penetrant than fesoterodine. While fesoterodine may have greater CNS penetration than Cobenfy's trospium, MPLT has done extensive work here to characterize/optimize the right dosing

And Much Higher Quantities Of Feso Are Needed To Cancel Out ML-007's Effects In Mice

Amphetamine-Induced Hyperlocomotion



ML-007 reduces hyperlocomotion in mice by acting on brain M1/M4 receptors. To test the degree of CNS penetration of fesoterodine, MPLT continued increasing feso doses until the effect of ML-007 starts to wane. Small impacts on hyperlocomotion were observed only at ML-007:PAC ratios around 10:1 to 20:1, which is far more fesoterodine than the 100:1 to 600:1 they are aiming for in human studies.

Furthermore, Between Fesoterodine and Trospium, the Latter Showed More CNS Anticholinergic AEs in Real-World

In addition to MPLT's mouse data, the comparison of CNS anticholinergic effects bolsters our confidence that fesoterodine isn't impacting the CNS as much as trospium. Based on post-marketing studies, trospium results in seemingly more CNS anticholinergic AEs (i.e., confusion, hallucinations) than fesoterodine at doses similar to Cobenfy (up to 125/30mg BID) and ML-007C-MA (210/3mg BID or 330/6mg QD). Furthermore, when determining the optimal ML-007:PAC ratio, mgmt. was aiming for more pro-cholinergic than anti-cholinergic activity which is reflected in AEs observed and notable absence of constipation and dry eye, for example. Bottom line: though fesoterodine can in theory enter the CNS, we think the overall balance still leans heavily towards pro-cholinergic activity in the brain.

Trospium Has More CNS Adverse Events Than Fesoterodine In Practice

Drug	Dosing	CNS Adverse Events on Label
Sanctura (trospium chloride)	20mg BID	Dizziness, confusion, hallucinations and somnolence have been reported in post-marketing studies
Toviaz (fesoterodine fumarate)	4 or 8mg QD	Headache, dizziness, and somnolence have been reported in post-marketing studies

Labels for trospium and fesoterodine monotox both contain warnings on CNS adverse events. These were observed in post-marketing studies, so it's impossible to be sure these are drug-related, we do note that trospium CNS AEs, which includes confusion and hallucinations, seem more severe than fesoterodine, which suggests the former may have more CNS anticholinergic activity.

MPLT Intentionally Skewed Towards Procholinergic AEs

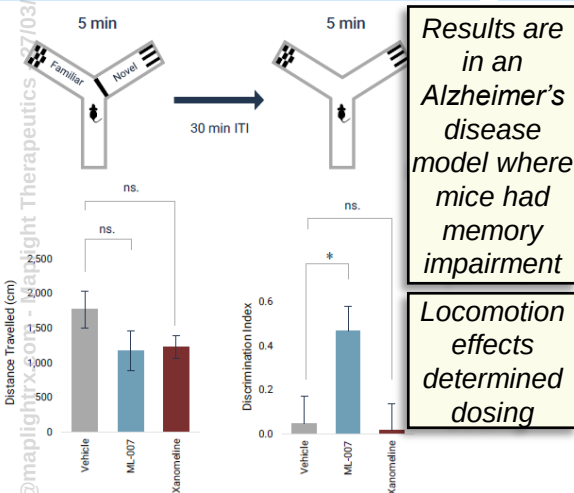
	Placebo (n=8)	210/3mg BID (n=6)	330/6mg QD (n=6)
Most common TEAEs (>1 subj)			
Nausea	0%	50%	50%
Dizziness	0%	50%	33%
Chills	0%	0%	33%
Hyperhidrosis	0%	17%	33%
Dyspepsia	0%	0%	33%

* No constipation or vomiting events were reported at these doses

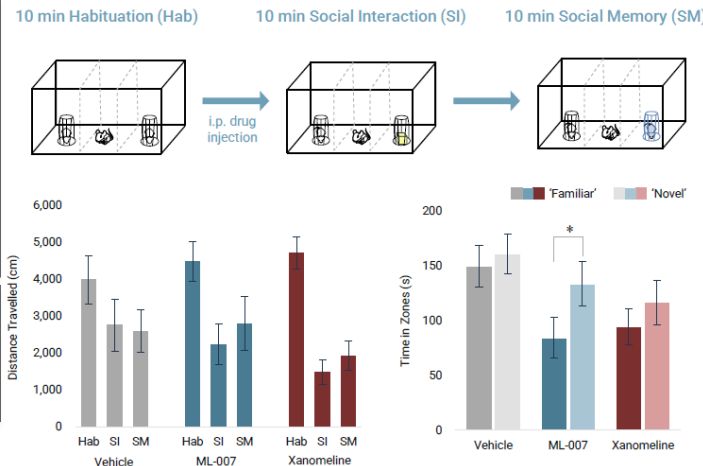
With Obvious Caveats, MapLight Has Also Done Head-to-Head Preclinical Work Suggesting They May Have Greater Cognitive Benefits Over Xanomeline

In a mouse model of Alzheimer's disease, ML-007 showed a statistically significant benefit on spatial and social memory—xanomeline did not demonstrate these same benefits at an equipotent dose which was determined by each drug exhibiting similar effects on locomotion (3mg/kg for xanomeline and 0.3mg/kg for ML-007). Management has highlighted how this pro-cognitive benefit stems '007's M1 bias/stronger agonism. We also remember old LLY literature describing xanomeline as having ~10x greater intrinsic activity at M4 vs M1. Again, the bottom line here is this is animal data and these models have questionable translatability. But it's another point of evidence that greater activity could translate into a differentiated effect – and the cognition angle becomes even more interesting in ADP.

Spatial Memory: Y-Maze



Social Memory Model



Results in Summary

✓ In both memory tasks, ML-007 showed statistically significant improvements where xanomeline did not in age-matched mouse cohorts. MapLight believes this pro-cognitive benefit stems from stronger M1 agonism.

✗ Xanomeline also targets M1, so we were a little surprised to see no benefit over vehicle on spatial memory. We do know though that Cobenfy is more active at M4 vs M1. And animal models for cognition always come with limitations.

*Significantly different from vehicle (p < 0.05)

Source: KRTX Materials; MapLight Materials; Stifel Research.

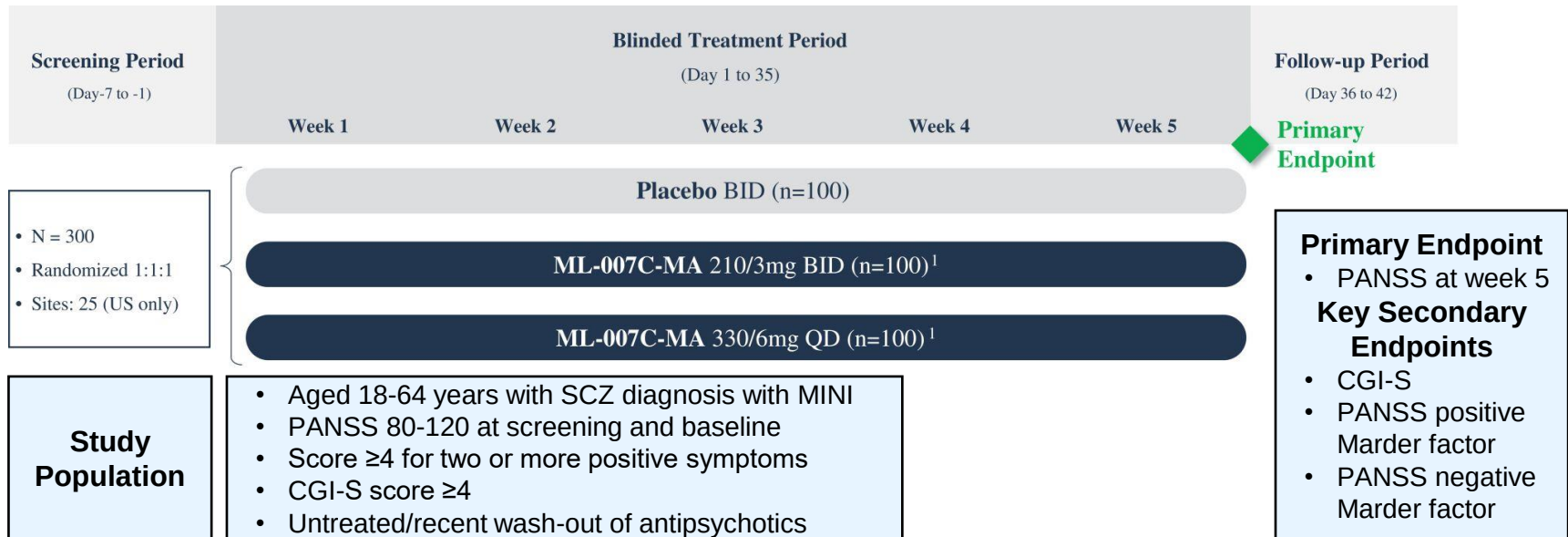
Lastly, Ease of Use Plays a Big Role in Antipsychotic Selection--MPLT's Improvements Over Cobenfy are Generally Perceived as Meaningful By KOLs

While “convenience” tends to be less of a priority than efficacy and safety, its benefits to commercial adoption cannot be overlooked especially in schizophrenia where compliance is a problem. On this aspect, MPLT potentially offers several advantages over Cobenfy, including (1) less frequent dosing, (2) ability to take the medication with meals, as opposed to requiring fasting, and (3) shorter dose titration period. On points #1 and #2, KOL feedback has verified that these are current limitations with Cobenfy, though they don't view titration as an issue (some psychs even like this). But in all, these incremental improvements help cement MPLT's competitive advantage over Cobenfy in schizophrenia even if the clinical data were roughly equal. Furthermore, the convenience is further appreciable in ADP, where Cobenfy's TID dosing is more arduous (discussed later) and may even be a hinderance to maximizing efficacy.

	ML-007C-MA	Cobenfy	KOL Feedback
 Dosing Interval	Potentially QD for schizophrenia, BID for ADP	- BID for schizophrenia - Current pivotal studies in ADP use TID dosing	“Half of my patients would find a new BID med burdensome--others already have other BID meds.” “Patient needs to be slightly more motivated to take a BID over QD, and it's helpful to have caregivers to remind them to take their doses.”
 Food Interaction	- Can be administered with meals - Fat content of meal does not influence absorption	1 hr before meal or 2 hrs after meal	“Timing dosing of Cobenfy around food can be challenging. One of my patients was missing half their doses because of their meal timing.” “Compliance will improve if patients don't have to time their doses around their meals.”
 Titration	- For schizophrenia, only a single lower dose for titration 1 week titration for ADP	1 week titration for schizophrenia - ADP likely uses even longer titration period	“Generally, I would be cautious since I prefer more approved doses so I can customize for those not tolerating max dose.” “I do an even slower titration than the Cobenfy label. It's better to go slow and acclimate patients vs a rigid titration.”

MapLight Is Currently Running a Phase 2 Study In Schizophrenia Patients That Is Set to Have Topline Data 2H26

The ZEPHYR Phase 2 Trial Design

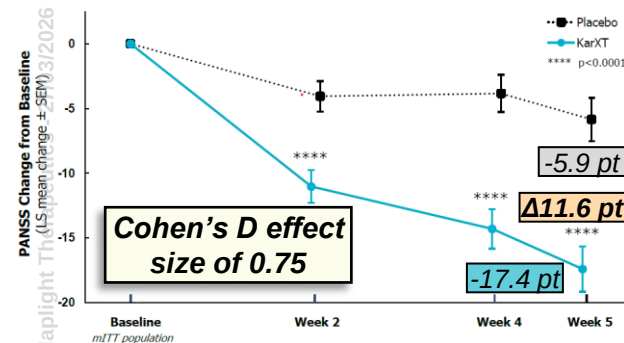


MapLight recently initiated a phase 2 study of their M1/M4 agonist ML-007C-MA in schizophrenia. This trial builds on the ph1 results in HVs and will explore both once and twice daily dosing. The design here is similar to Cobenfy's ph2 (EMERGENT-1), including the primary endpoint of change in PANSS from baseline to week 5. MapLight is aiming to enroll ~300 patients in the US (KRTX enrolled 182 patients across 2 arms), which could make this study big enough to count as a pivotal trial

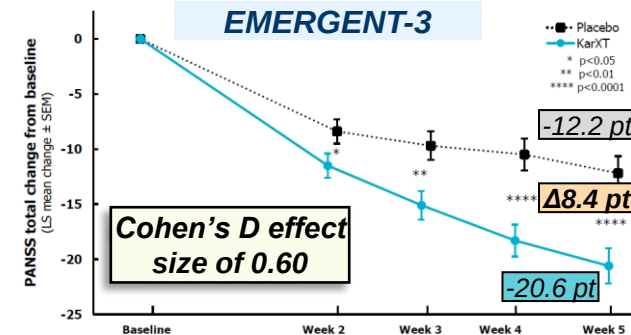
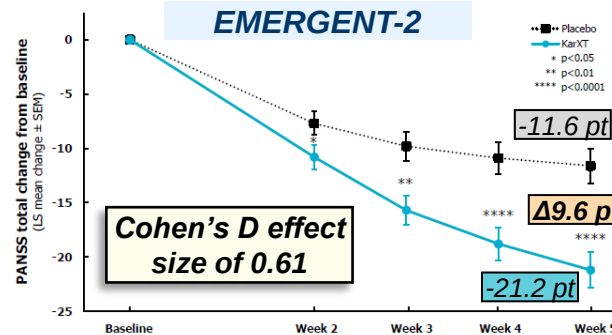
As an Analog, Cobenfy Saw Robust Ph2 Results That Translated to Ph3 Success

Recall in ph2, Cobenfy saw a robust effect size (Cohen's d of 0.75) that was at the top end of all antipsychotics. This result was somewhat aided by a lower than typical placebo response (5.9-point reduction), but encouragingly, Cobenfy only saw a slight regression in the ph3 effect size (Cohen's d = 0.6). While Cobenfy sets a high bar here, we think MapLight is set up well to show similar results given the large trial design (~100 patients per arm), with the caveat that with all the buzz around muscarinics we may see greater placebo effects. Again though, showing better efficacy is not the "bar" – it's really about establishing '007 as an effective drug with some improvements on dosing/tolerability.

Robust Efficacy in Ph2



Which Translated into Ph3 Results



In ph2, Cobenfy demonstrated an 11.6-pt placebo-adjusted improvement on PANSS total score—this was somewhat aided by a lower placebo effect, the overall reduction on drug was significant and held up in ph3.

Ultimately in ph3, Cobenfy showed a very robust and highly significant reduction in PANSS. While the effect size came down a bit from ph2, these results continued to represent efficacy that is above average for antipsychotics. Admittedly, this sets a high bar, but we still think there's room to differentiate on efficacy though we think in SCZ, other aspects like ease-of-use and side effect profile can often drive meaningful use.

In General, Muscarinic Tolerability Seems Similar Between HV and Schizophrenia Patients—We Saw This for Cobenfy, and It Should be the Same for ML-007

As an Analog: Cobenfy Saw Similar Safety in SCZ Patients vs HVs

	Ph1 POC		Ph1 SAD			Ph2: EMERGENT-1		Ph3: EMERGENT-2		Ph3: EMERGENT-3	
	X + PBO	Cobenfy	PBO	100mg X /20mg T	125mg X /40mg T	PBO	Cobenfy	PBO	Cobenfy	PBO	Cobenfy
Total cholinergic AEs	64%	34%	24%	39%	39%	-	-	-	-	-	-
Nausea	24%	17%	-	-	-	4.4%	16.9%	6%	19%	1.6%	19.2%
Vomiting	15%	6%	-	-	-	4.4%	9.0%	1%	14%	0.8%	16.0%
Diarrhea	21%	6%	-	-	-	-	-	3%	6%	0.8%	5.6%
Sweating	49%	20%	-	-	-	-	-	-	-	-	-
Salivation	36%	26%	-	-	-	-	-	-	-	-	-

X=xanomeline; T=trospium.

While safety details in ph1 development were relatively limited, Cobenfy tolerability in schizophrenia patients vs in healthy volunteers was similar in terms of cholinergic side effects. This is different from dopaminergic antipsychotics that tend to be better tolerated in schizophrenia patients, but the disparity may just be due to biological differences.

Commercial Precedent in Psychiatry Shows That the Bar for Differentiation Is Low—Incremental Benefits Can Often Justify Meaningful Use, Significant Sales

Many newer drugs in psychiatry gain usage through very incremental improvements—in the examples below of more recent psychiatry products, often all that's needed is a slight improvement on tolerability to justify meaningful use. As such, we think that if ML-007C-MA were to show a similar risk-benefit profile to Cobenfy but with improved ease of use, that would be enough for the space to consider this a differentiated product, commercially speaking.

Differentiation from SOC:

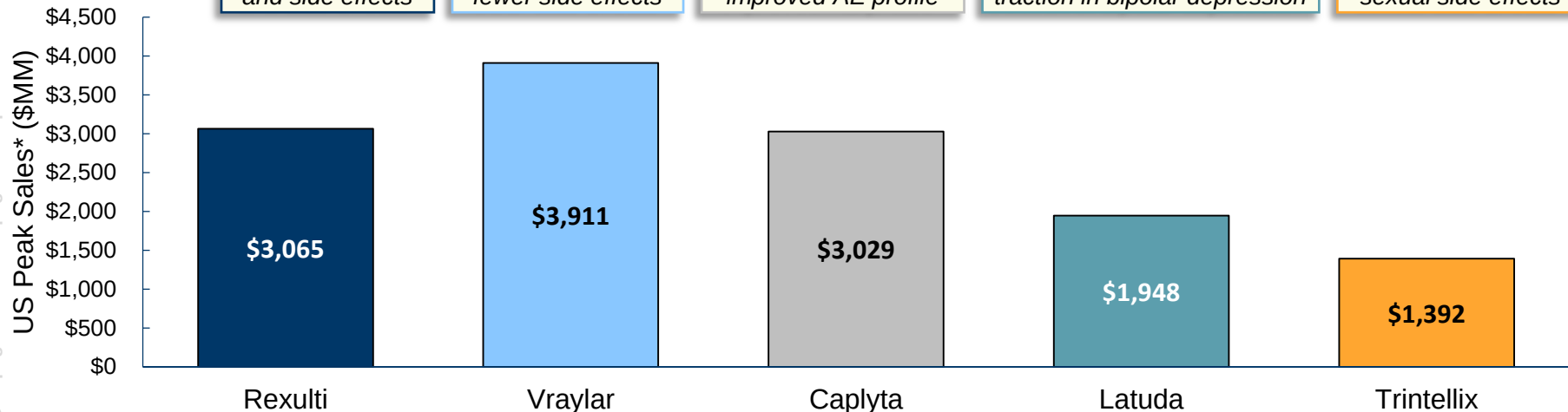
Rexulti provides an incremental benefit over Abilify in regards to dosing and side effects

Vraylar has differentiated pharmacology that results in slightly fewer side effects

Despite seeing less PANSS efficacy, Caplyta has garnered use through an overall improved AE profile

Like Caplyta, Latuda has more modest efficacy but improved tolerability—this has given way to Latuda gaining greater traction in bipolar depression

Trintellix is a MAO inhibitor that is differentiated in MDD by having lesser sexual side effects



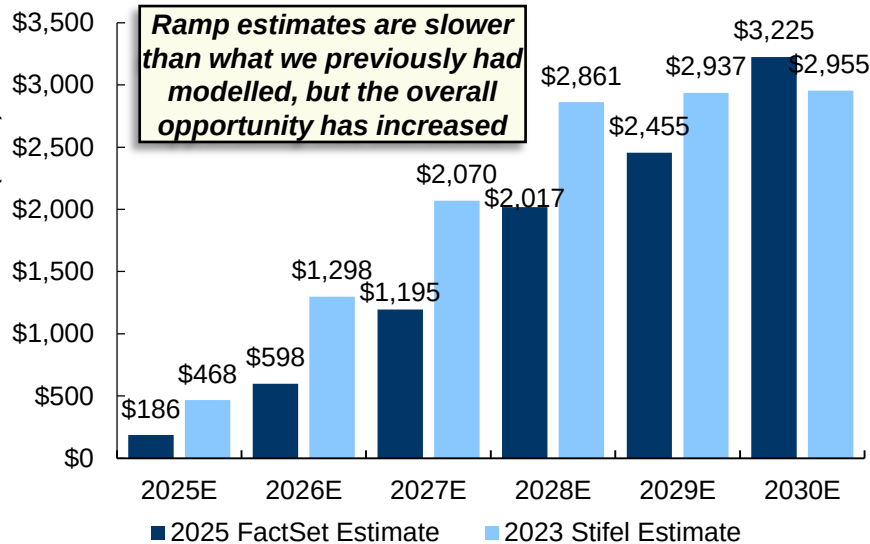
*All values are estimates except for Latuda

Source: EvaluatePharma; MapLight Materials; Stifel Research.

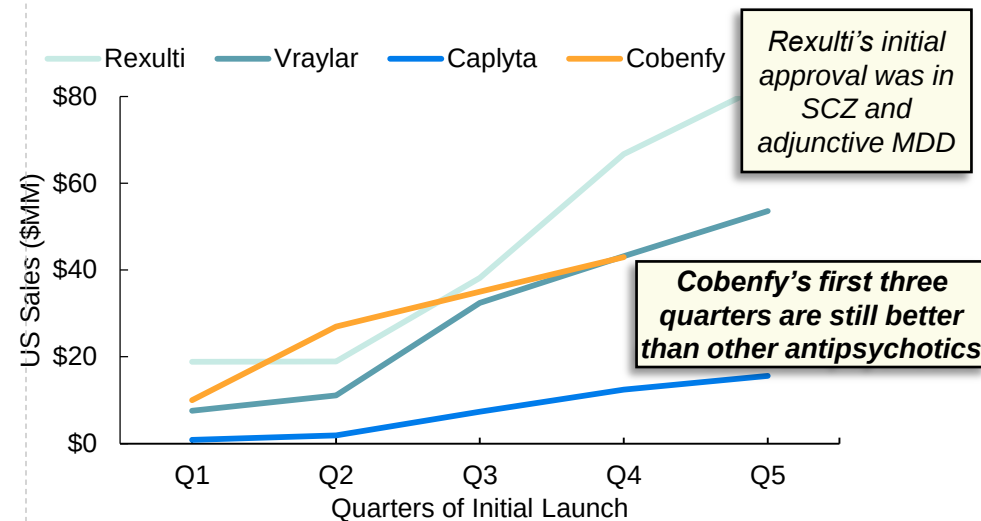
One Bear Case Is That Cobenfy's Launch In Schizophrenia Has Been Slower Than Many Expected – We Still See the Market as Big Though

One bear case here is that there's already a number of antipsychotics on the market and recent schizophrenia launches have been not so great. Cobenfy was supposed to break this precedent, and while it is doing better than recent atypical launches, the drug is tracking below where Karuna consensus was. As we'll get to later, the schizophrenia branded market opportunity still remains very real and KOL feedback remains supportive. And as it relates to the Cobenfy adoption curve, there are probably issues specific to schizophrenia that make drugs launch slower at first: (1) demand isn't really patient driven – vs say MDD/bipolar where patients may come in asking for a drug, (2) a lot of NRx in schizophrenia come in the community or the acute treatment setting, and changing prescribing behavior here likely takes time. That said, KOLs we've spoken to are still confident that Cobenfy/muscarinics will take a large share over time – the launch may just be more linear than previously thought.

Consensus Is Now Less Than Our Initial Estimates



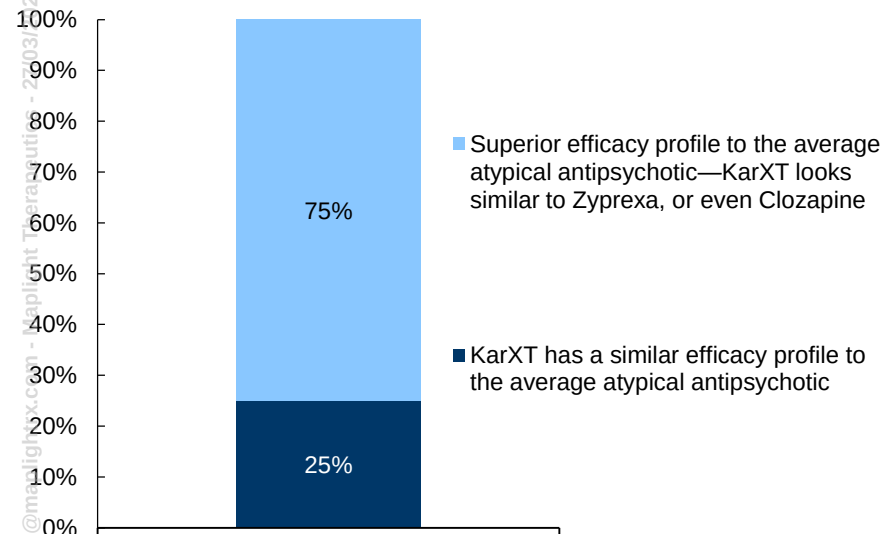
But in Reality, the First Few Quarters Looked Ok



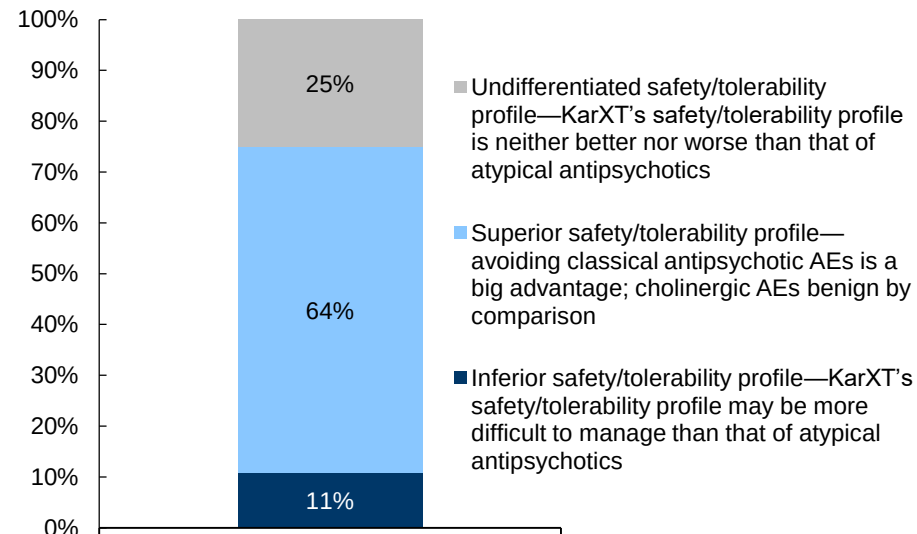
Our Prior Survey of 28 Doctors Was Very Bullish on Cobenfy Utilization Prospects

Pre-PDUFA, we conducted a survey of 28 psychiatrists who highlighted enthusiasm for Cobenfy's efficacy and safety/tolerability profile vs the schizophrenia standard-of-care. With the caveat that drugs often survey better than they sell, we think this feedback here is bullish for MapLight and the muscarinic opportunity broadly.

KarXT Efficacy vs Schizophrenia Standard-of Care



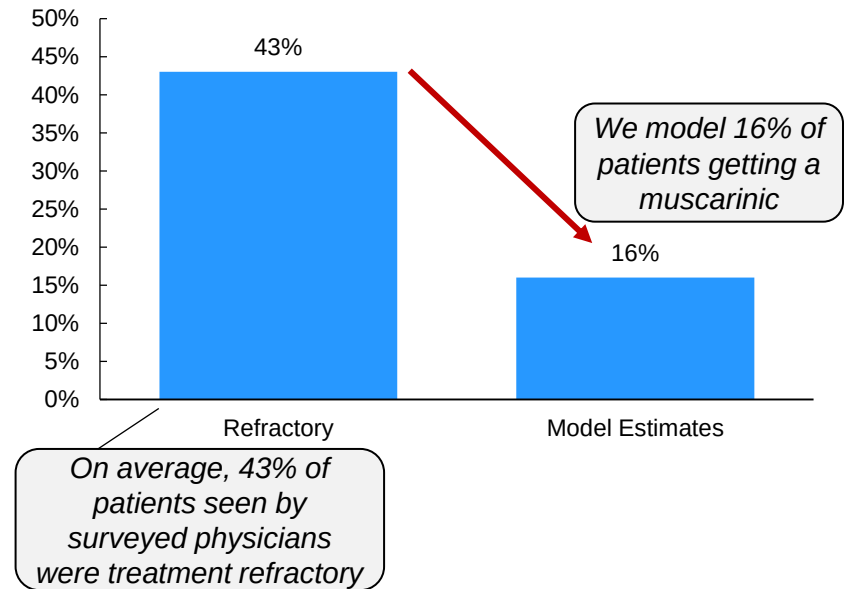
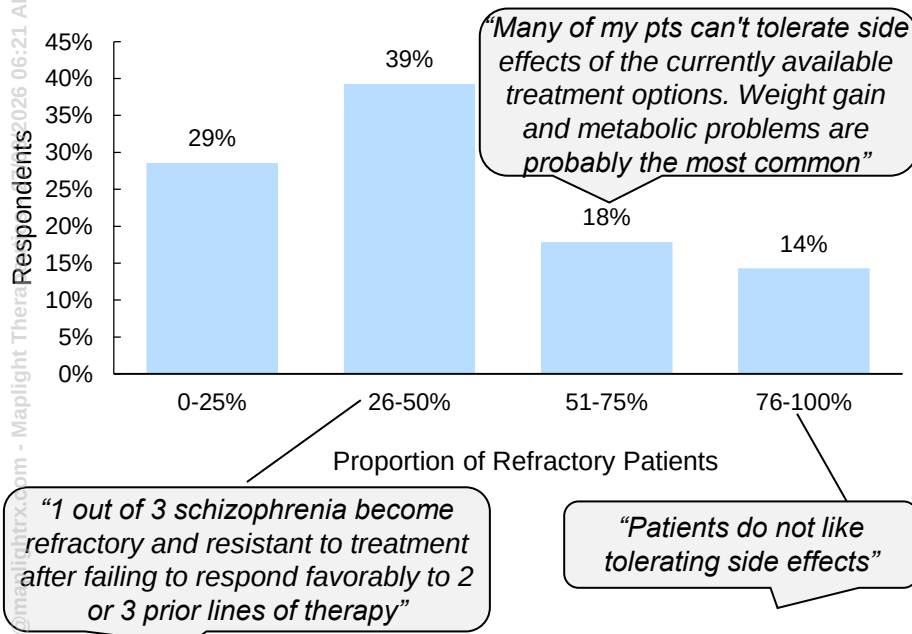
KarXT Safety vs Schizophrenia Standard-of Care



Source: MapLight Materials; Stifel Research

Surveyed Physicians Estimate That 45% of Schizophrenia Patients “Inadequately” Respond to Standard of Care – That’s a Large Opportunity

Responders estimated that 45% of schizophrenia patients have an inadequate response to SOC antipsychotics—this is materially higher than the ~30% treatment refractory number we typically see in the literature. Further, physicians highlighted how tolerability is a major challenge for patients. In our model, we discounted this data because almost all drugs survey better than they sell. But at face this implied a >\$10B opportunity for muscarinics – so even with a very significant haircut (i.e. 50%+) because drugs often survey better than they sell, the market here can still be really significant.



Real-world Experience with Cobenfy has been Generally Positive So Far, with Prescribers Noting Strong Patient Satisfaction

With Cobenfy available on the market for almost a year, we recently gathered physicians' perspectives on real-world experience. Our key takeaways were that (1) efficacy was perceived to be comparable to other antipsychotics, (2) safety and tolerability are better than expected from clinical trials as physicians slowly titrate, (3) many patients on a muscarinic felt subjectively better than on a dopaminergic, and (4) one challenge is uptake may be dampened by high OOP costs for Cobenfy vs generic antipsychotics. Bottom line though – experience has been generally good and the interest in a novel MOA is high.

Real-World Cobenfy Experience



Efficacy

- “While it’s not on the same level as clozapine, I’d say the efficacy is pretty competitive, similar to risperdone or olanzapine”
- “I find its efficacy is not as good as lurasidone, though it has impressive safety/tolerability and other improvements”



Safety/ Tolerability

- “The initial GI symptoms are expected and mostly short-lived, but with prophylactic antiemetics and patient education beforehand, none of my patients have discontinued due to these adverse events”
- “Patients on dopaminergic tend to be on heavier side due to metabolic side effects, so a bit of nausea isn’t the worst thing. Constipation is the main concern especially if persistent”



Patient Satisfaction

- “Patients really like it. I’ve had 2 pts who didn’t respond so we stopped Cobenfy, but they are asking to try again-- being on a dopaminergic just feels worse”
- “My patients prefer the GI side effects over the dopaminergics which make them groggy.”



Access

- “Compared to generic dopaminergic antipsychotics, Cobenfy is fairly expensive, so not all patients are willing to pay the out-of-pocket costs.”
- “Usually there’s a prior auth, but even if that was approved patients could still pay \$200/month on copay.”

BMJ's Survey Further Supports Docs' Positive Reception to Cobenfy, Though GI AEs are Difficult To Manage Even Despite Slower Titration

BMJ surveyed psychiatric HCPs to understand their rationale / experience with starting patients on Cobenfy and received n=90 responses, each representing a unique patient. Of note, while these patients were commonly switching from either olanzapine (n=22) or risperidone (n=17), some were antipsychotic-naïve altogether (n=5) which is promising but could just be sampling bias. Efficacy was the primary reason for discontinuing the prior antipsychotic, though safety (metabolic side effects, EPS) were also noteworthy drivers. Interestingly, a meaningful number also believe in Cobenfy's ability to improve cognition, even if this was only in a subgroup analysis and not on the formal label. Meanwhile, despite slower titration, nausea and vomiting are still prominent and only partially mitigated with either ondansetron or additional tropism--MPLT could leverage this weakness if they can deliver on safety/tolerability.

Reasons for Discontinuing Dopaminergic

Reason, %	Adults With Schizophrenia (N=90)
Lack of adequate control of positive symptoms (hallucinations, delusions)	46.7
Worsened or lack of improvement of negative symptoms (apathy, anhedonia, social withdrawal)	46.7
Weight gain and metabolic changes (obesity, diabetes, dyslipidemia)	34.4
Cognitive dulling or emotional blunting	34.4
Sedation or fatigue that interferes with daily functioning	23.3
Emotional flattening	23.3
EPS (tremor, rigidity, akathisia, dystonia)	13.3
Limits motivation or creativity	11.1
Interferes with work, school, or social life	10.0

Majority of patients were started on Cobenfy because the prescriber believed its orthogonal mechanism should translate into efficacy. However, the distinct AE profile was also a key driver, with many wanting to avoid the metabolic changes and EPS associated with dopamine-targeting antipsychotics.

Reasons for Initiating Muscarinic

Reason, %	Adults With Schizophrenia (N=90)
Novel MOA	74.4
Improvement in cognitive impairment	70.0
Failure of multiple prior D ₂ dopamine receptor blocking antipsychotics	64.4
Improvement in mood symptoms and negative symptoms	64.4
Lack of weight gain or metabolic side effects	61.1
Reduced EPS or tardive dyskinesia	58.9
Efficacy across domains of symptoms of schizophrenia	57.8

Tolerability Seems to be an Issue

Figure 4. Maximum dose of X/T administered

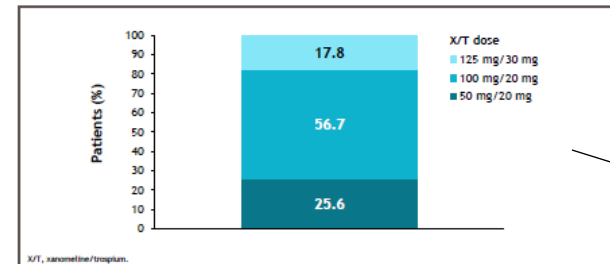
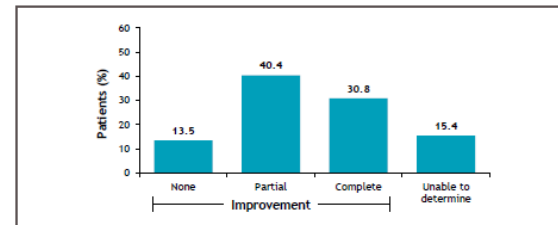


Figure 5. Effect of pharmacologic treatment on nausea and vomiting



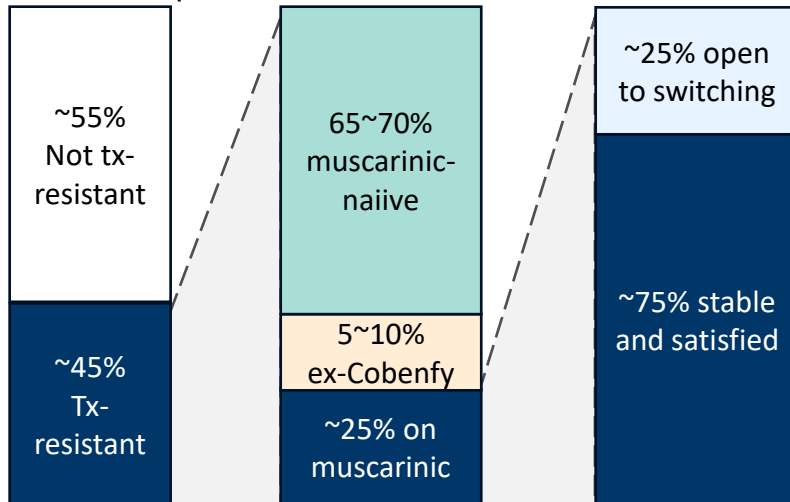
Most of the patients underwent a longer titration scheme, with ~60% staying on the lowest dose (50mg/20mg) for up to 2 weeks. Despite that, 64% and 37% experienced nausea and vomiting respectively, of which ~55% received pharmacological intervention with mixed success.

If MPLT Achieves Its Target Product Profile (QD), They Should Be Well-Positioned To Capture Share Within The Muscarinic Class Over Time

If Maplight were able to garner FDA approval with once daily dosing and no fasting requirements, even with no additional benefits on efficacy or safety (and we are hopeful tolerability will improve), docs may preferentially prescribe ML-007C-MA over Cobenfy. Despite Cobenfy's lead to market, we foresee several segments of the schizophrenia population that MPLT would specifically appeal to: (1) muscarinic-naïve treatment-resistant pts, (2) those that previously tried Cobenfy and saw some initial efficacy but discontinued due to lack of efficacy or AEs, and (3) those stable on Cobenfy but would prefer more convenient dosing. In schizophrenia the average duration of treatment is historically short (less than a year), so with lots of cycling a second mover drug can gain meaningful share.

Stifel Estimate Of Schizophrenia Market Segments In 2030

3MM US SCZ pts



MPLT vs Cobenfy For Different Segments

Muscarinic-Naïve Patients

"No reason not to start with MPLT first, just because it's easier to use than Cobenfy"

Ex-Cobenfy Patients

"Of the 8 patients I've started on Cobenfy, 2 have discontinued due to suboptimal efficacy"

"I've had 3 out of 16 discontinue on Cobenfy, 1 due to failed efficacy (relapse), 1 due to GI intolerance, and 1 with both"

Switching From Cobenfy

"Wouldn't want to rock the boat if they're doing well, but some may be missing doses"

"It'll be a small fraction, maybe 1 in 4 Cobenfy users that have suboptimal adherence."

While Schizophrenia Is a Meaningful Market, Potential Success In ADP Is a Swing Factor That Could Increase the Commercial Opportunity Multifold

Bottom line: schizophrenia remains a compelling market opportunity for muscarinics that we expect to grow significantly over time, even if the initial rollout is slower than expected. That said the Alzheimer's population—where there are no approved drugs to treat psychosis and very few to treat behavioral symptoms more broadly—is a significant market that represents an ideal label-expansion opportunity, as exemplified by Rexulti which recently garnered approval in AD agitation that led to significant inflection in TRx growth. There's a huge unmet need here as atypical antipsychotics have sub-optimal efficacy, and a black box warning for an increased risk of mortality. Prior KOL feedback suggests that garnering an ADP label indication is one of the clearer paths to having a blockbuster antipsychotic.

ADP: A Growing and Untapped Market

Schizophrenia

~3MM patients

Alzheimer's
Disease
Psychosis

6-7MM Alzheimer's
patients

20-40%

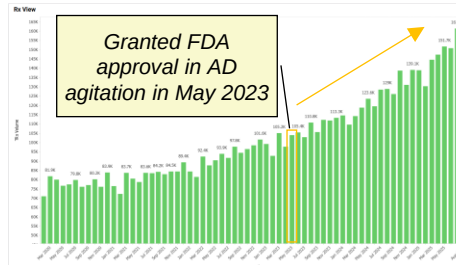
1.2-2.8MM ADP
patients

Unlike schizophrenia where US prevalence estimates are fairly narrow, there's a wider estimate for prevalent ADP patients

We note: (1) estimates for ADP are imprecise, but even the low end is a big market, (2) as the US population continues to age, the prevalence of Alzheimer's and thus ADP may rise significantly over the next two decades, and (3) the ADP estimates are only for patients that experience delusions/hallucinations [i.e., not including aggression/agitation]. And perhaps most importantly—as we will get into—while there are numerous treatment options for schizophrenia, there are no FDA-approved agents for ADP and atypicals have a black box warning for morbidity/mortality in dementia pts

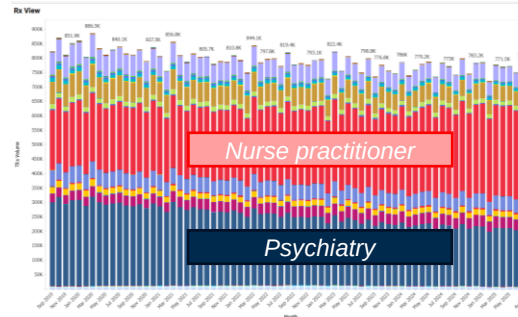
Rexulti, Risperidone TRx Data Exemplify Oppty in AD

Rexulti's Recent Inflection in TRx Volume Driven by Use in AD



Rexulti (brexpiprazole) first was granted FDA approval for both schizophrenia and adjunctive MDD in July 2015. Script volumes were more steady recently and then began to meaningfully inflect following positive late-stage data in AD agitation in 2022—we view it as very likely the majority of the script growth (+60% in a little >2yrs) can be attributed to use in Alzheimer's

Risperidone TRx By Specialty May Suggest Greater Use in Dementia



IQVIA data shows that while overall TRx volume for risperidone may have experienced only a modest decline (~7%) since 2020, the % of overall scripts written by psychiatrists has declined by ~25%, while nurse practitioners have taken greater share of overall scripts written. This suggests a greater proportion of TRxs are being written off-label to treat behavioral symptoms of dementia

History Of Antipsychotics In ADP: Often Used Despite Very Mixed Data/Safety, Including A Class Black Box Warning for Mortality Risk

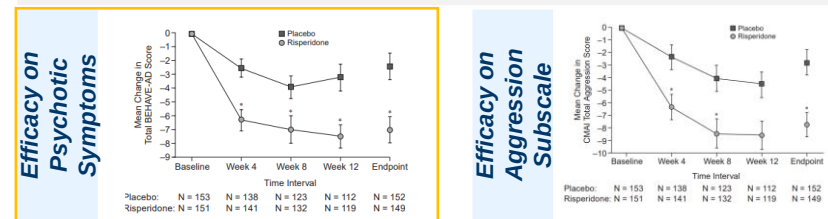
While there are currently no FDA approved drugs indicated for ADP, a meaningful proportion of patients today are being treated with low-dose, second-gen atypical antipsychotics. This practice has persisted despite the fact that in 2005, the FDA issued a black box warning that for treatment of behavioral symptoms in patients with dementia, second-generation antipsychotics (e.g., risperidone, quetiapine, aripiprazole, brexpiprazole) were associated with increased morbidity and mortality, data supported by independent meta-analyses. While brexpiprazole garnered approval in AD agitation, none have been able to garner approval in ADP. Risperidone has demonstrated some evidence of benefit on cognition (and aggressive symptoms, which they received approval for specifically, but only in the EU), data from others in psychosis is mixed at best.

Atypical Antipsychotics A Mainstay of Tx in AD Psychosis, Aggression

Next-generation atypical antipsychotics, characterized by their mechanism of inhibiting activity of D2 (dopamine) and 5-HT2A (serotonin) receptors in the brain, have proven efficacious in treating positive symptoms of schizophrenia, for which several therapies have received FDA approval. These therapies have also garnered meaningful use outside of schizophrenia, with several having approvals in bipolar disorder, major depression and other psychiatric indications. **Further, off-label use of atypical antipsychotics to treat AD psychosis and aggression is highly common, despite a black box warning placed on this class of agents for increased mortality risk in dementia patients. This underscores the significant unmet need and opportunity in ADP, where there are no specific FDA approvals.**

Risperidone Efficacy Demonstrated in AD Psychosis Distinguishes It From Other Atypicals

Risperidone was assessed in patients with AD/dementia w/ symptoms aggression in a 12-wk, DB, pbo-controlled RCT in a large sample (N=345), where it demonstrated success on aggression and psychotic symptoms. Success here supported EU approval for short-term use in AD aggression



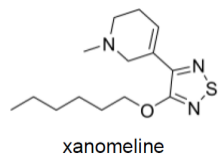
Risperidone demonstrated stat. sig. differences vs. PBO on aggression subscale of CMAI ($p < .001$), as well as BEHAVE-AD, a scale for psychotic symptoms ($p = 0.003$). Efficacy here is among the best demonstrated by atypical antipsychotics, and risperidone has been able to show more consistent efficacy in treating symptoms of aggression

Xanomeline Demonstrating Efficacy in A Ph2 Study In AD Set the Stage for Development of KarXT (And Now ML-007-MA) in ADP

Xanomeline—a M1/M4 receptor agonist and key component of KarXT—notably was originally developed by Lilly for Alzheimer's where it demonstrated efficacy on cognitive scales in a ph2 study, an important finding that validated exploration of muscarinics in this indication. Moreover, there was a surprise finding in a post-hoc analysis, where LLY observed a dose dependent beneficial impact on the frequency of both hallucinations and delusions. Xanomeline alone was extremely difficult for ADP patients to tolerate, but it laid the groundwork for KarXT in this indication, where we now await data soon from the first Alzheimer's ph3.

Xanomeline: A M1/M4 Preferring Muscarinic Agonist, Originally Developed by LLY for Alzheimer's

Originally developed as part of Eli Lilly's Alzheimer's program, xanomeline—one of the components of KarXT/Cobenfy—is a preferential M1/M4 muscarinic agonist. M1 activation is believed to be associated with improvement in cognition, whereas M4 activation has been hypothesized to be the primary driver of antipsychotic benefit. You never totally know though...



Receptor	EC ₅₀ (nM)
M ₁	7.6
M ₂	125
M ₃	67
M ₄	20
M ₅	508

Structure and selectivity of xanomeline

Xanomeline Demonstrated Efficacy on Cognition and Psychosis AEs in Alzheimer's Patients

High-dose xanomeline demonstrated statistically significant improvement vs. PBO on cognitive scales (ADAS-Cog, CIBIC+) in a large ph2 study in Alzheimer's (i.e., "probably AD"). There was also a compelling decrease in the rate of hallucinations and delusions. The discontinuation rate for the high dose though was 50%, largely due to nausea/vomiting, and LLY also saw an imbalance in syncope.

Source: Bodick. 1997.

Xanomeline's AD Signal Drove Interest in Potential of Muscarinics, Led To Dev't of KarXT in ADP

Xanomeline's signal here led KRTX to conduct significant ph1 work to optimize the KarXT dose in ADP. BMY is running three ph3 studies in parallel in ADP—one is expected to readout imminently (by YE25), while the remaining two are expected to readout next year

Indication	Psychosis in Alzheimer's Disease (ADP)		
	Phase III - ADEPT-1	Phase III - ADEPT-2	Phase III - ADEPT-4
# of Patients	N = 380	N = 400	N = 406
Design	- Cobenfy 20 mg/2 mg TID, 30 mg/3 mg TID, 40 mg/4 mg TID, 50 mg/5 mg TID, 60 mg/6 mg TID - Placebo	- Cobenfy 20 mg/2 mg TID, 30 mg/3 mg TID, 40 mg/4 mg TID, 50 mg/5 mg TID, 60 mg/6 mg TID - Placebo	- Cobenfy 20 mg/2 mg TID, 30 mg/3 mg TID, 40 mg/4 mg TID, 50 mg/5 mg TID, 60 mg/6 mg TID - Placebo
Endpoints	- Primary: Time from randomization to relapse during the 26-week double blind randomized withdrawal period - Key secondary: Time from randomization to withdrawal for any reason during the 26-week Double-Blind Randomized Withdrawal treatment period	- Primary: Change from baseline in Neuropsychiatric Inventory-Clinician: Hallucinations and Delusions (NPI-C-H-D) score to end of Week 14 - Key secondary: Change from baseline in Clinical Global Impressions-Severity (CGI-S) scale up to week 14	- Primary: Change from baseline in Neuropsychiatric Inventory-Clinician: Hallucinations and Delusions (NPI-C-H-D) score up to Week 14 - Key secondary: Change from baseline in Clinical Global Impressions-Severity (CGI-S) scale up to week 14
Status	- Recruiting - Projected data readout 2026	Projected data readout 2H 2025	- Recruiting - Projected data readout 2026
CT Identifier	NCT00511363	NCT01248224	NCT01585787

Investor Expectations for Cobenfy In the First Ph3 ADP Readout (ADEPT-2) Are Low; To Understand MPLT Implications We Will Need the Full Data

Naturally, we expect investors to be paying close attention to the outcome of Cobenfy's ADP program, which is currently in phase 3 and expecting the first of multiple readouts by the end of this year (ADEPT-2). Generally speaking, expectations are lower on this first ph3 readout: BMJ has referred to conducting clinical trial site reviews prior to readout (suggesting that there may have been a conduct issue) and further, the delays to the readout coupled with the higher-risk profile of the program has driven some skepticism here. While we'd view success here (and for the program more broadly) as a clear win for MPLT that should send the stock a lot higher, a failure could be negative but it also could be nuanced – for example a clear efficacy trend but a study that failed due to compliance issues, dropouts or patients not being able to titrate all the way up to the highest dose would arguably lay out a potential path to success for MPLT

Challenges with Execution

We've seen examples of a company getting acquired and subsequently fail late-stage studies—most notably, competitor muscarinic company CERE (acquired by ABBV) failed in its pivotal schizophrenia studies. Changes in management can lead to oversight on execution, which is especially important in psychiatry studies where patient selection and managing placebo response are so critical. BMJ has alluded to this potentially being a factor, noting they're conducting a trial site review prior to disclosing the results

"Specifically as it relates to ADEPT-2, in preparation of that database lock, we are conducting the clinical trial site reviews that could impact the timelines a bit." -- BMJ CMO, July 2025)

Poorer Tolerability in Elderly Patients

One of the key challenges of this MOA / drug target is challenges associated with tolerability, and we believe this is even more pronounced in an elderly population with Alzheimer's, who are particularly sensitive to side effects. This has been reflected in both BMJ/KRTX and MPLT conducting extensive ph1 work here with healthy elderly to inform dosing, which for BMJ/KRTX was TID dosing with a much slower titration. Also, we saw more detailed healthy elderly for KarXT in the FDA review docs, and AE rates were much higher. And while we think it's still early days for MPLT showing differentiation on tolerability in schizophrenia, in ADP, because MPLT is BID (not TID) and lacks a food issue, one could argue that a KarXT failure (depending on details) shouldn't imply that ML-007 will necessarily fail too.

Bottom line: we'll have to see what the data show, but as ADP is an area where the ML-007 profile might be relatively more differentiated, the ADEPT-2 data will need to be interpreted in context

Maplight Also Ran an Elderly Safety Cohort to Assess Tolerability Which Looks Fairly Consistent with Schizophrenia

Maplight has done significant safety work here, and in their 013 ph1 study in healthy elderly subjects, they assessed three distinct target doses (N=17) vs. PBO (N=7), leveraging a 2-7 day titration period followed by 7 days of dosing. MPLT identified a target dose (210/3mg) which was well tolerated—low rates of cholinergic AEs and no red flags. While the duration of the study is shorter with fewer patients than KarXT's trial (right), it appears MPLT's therapy is trending towards being a better tolerated drug in ADP (fewer discontinuations, etc).

MapLight's Drug Has A Clear Dosing Advantage vs. KarXT in ADP

Drug	Dosing	Food Effect
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 MapLight	ML-007C-MA	BID	No fasting required
 Bristol Myers Squibb®	KarXT	TID	Fasting required for max absorption

MPLT's target dose for ADP was identified following results from a ph1 study in healthy elderly patients (data on right) that supported BID dosing at 210/3mg. Overall, their dosing frequency (BID vs. TID) and ability for patients to dose without need for fasting are key advantages. MPLT has the potential to have only a 1wk titration (beginning at 50% of the target dose), whereas KarXT's may be lengthier, and starting at a ~1/3 of the max dose for the ADEPT-2 study.

ML-007C-MA Had Slightly Lower Rates of AEs, But Samples Are Small, Durations Are Limited and BMJ's Disclosure Was Limited

Study Design of KarXT Ph1 Elderly HV Study

KAR-030 (Phase 1)	Age 60 to 85 Healthy elderly	Control type: placebo-controlled Randomization: 3:1 randomized to placebo or KarXT Blinding: double-blind Innovative design	Drug: KarXT Dosage: Target TDD range from 150 mg/20 mg QD to 200 mg/40 mg (divided TID) Number treated: 36	Primary: PK, Safety	Randomized: KarXT: 36 Placebo: 12	3 United States
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	KarXT (N=36)	PBO (N=12)	ML-007C-MA (N=11)	Pooled PBO
% with ≥1 Moderate Drug-Related TEAE	47%	8%	7%	N/A
% with ≥1 Gr. 3+ AEs	3% ¹	0%	0%	0%
Constipation	50%	17%	N/A	N/A
Dyspepsia	47%	8%	N/A	N/A
Drymouth	36%	0%	N/A	N/A
Vomiting	19%	0%	N/A	N/A
Nausea	33%	0%	27%	0%
Dizziness	N/A	N/A	27%	14%
Hyperhidrosis	N/A	N/A	27%	0%
Headache	N/A	N/A	36%	0%
Discontinuation Rate	25%	0%	0%	0%

¹ Grade 3 urinary retention

Taking a Step Back, What's the Bear Case on MPLT Broadly? Anticipated Investor Pushback and Our Perspective

Potential Bear Arguments

- 1 *Pessimism on Potential for ML-007C-MA to differentiate from KarXT. Could this just be a KarXT me-too?*
- 2 *With Cobenfy sales off to a slower start than initially expected, is there a large enough commercial opportunity in schizophrenia?*
- 3 *If Cobenfy fails in ADP, how can we have confidence that MPLT will succeed?*

Stifel Perspective

- 1 *The totality of the preclinical/ph1 work suggests that (1) MPLT is a stronger M1/M4 agonist and (2) a potentially better tolerability profile on at least some AEs. And while it's possible that MPLT could fall short on differentiating in either efficacy/safety, we do think differentiation on dosing is most likely and arguably very meaningful commercially*
- 2 *Sales of Cobenfy through the first quarters of launch have still outperformed most of the other recent atypical antipsychotic drug launches (e.g., Vraylar, Caplyta) and we believe that muscarinics will be able to gain a meaningful share of this large market (~3MM patients US). KOL feedback still suggests a big market opportunity, albeit one that will take longer to develop.*
- 3 *If BMY reports a failed ADEPT-2 study in ADP, the readthrough may be difficult until we get more details and at face, many will probably interpret this as a clear negative. That said, BMY has already alluded to execution issues with the study (i.e. auditing sites) and we do think expectations are low to begin with. More broadly though, we think KarXT, as a TID drug that needs to be dosed fasted, may have more difficult in an ADP population and this could reduce POS. Conversely though, if ADP works for Cobenfy we think MPLT shares will likely trade substantially higher.*

Bottom line: we believe if MPLT were to succeed in SCZ, we believe they have a case to at least be somewhat differentiated from KarXT (particularly on ease-of-use/tolerability), and we still expect muscarinics to become a large class over time. Meanwhile, ADP is higher-risk and we can't say we have conviction – ADP is highly risk adjusted in our model. That said, we think the Cobenfy to ML-007 readthrough in ADP may be more nuanced, plausibly analogous to the BMY/CYTK situation in non-obstructive HCM.

There's Also Sources Of Optionality As BMJ Explores Cobenfy's Potential In Other Indications

BMJ is doing a lot of legwork to fully explore the potential of muscarinics beyond just schizophrenia and ADP. Since ML007C-MA is leveraging a similar M1/M4 mechanism, any potential successes from KarXT could also translate to MPLT. These are not credited in MPLT's current stock (or our valuation), but being a fast follower can be beneficial in prioritizing future indications.

Cobenfy Study	Indication	Clinicaltrials.gov Estimated Study Completion Date
Ph3 BALSAM-1	Mania in Bipolar I Disorder	Nov 2026
Ph3 BALSAM-2	Mania in Bipolar I Disorder	Nov 2026
Ph3 BALSAM-4	Mania in Bipolar I Disorder	TBD
Ph3 ADAGIO-1	AD Agitation	Dec 2028
Ph3 ADAGIO-2	AD Agitation	Nov 2028
Ph3 MINDSET 1	Cognitive Impairment in AD	Feb 2029
Ph3 MINDSET 2	Cognitive Impairment in AD	Feb 2029
TBD	Autism Spectrum Irritability	TBD

"We currently have 14 [Cobenfy] studies that are ongoing or in the process to be activated. 10 of those studies are pivotal studies." - BMJ CMO at 3Q25 Earnings Call

MPLT Is Also Evaluating Other Candidates In ASD, Which Is In Ph2, As Well As Earlier Stage Programs—All Are Higher Risk Sources of Optionality

Beyond the lead muscarinic program, MPLT is also advancing ML-004, a 5-HT_{1B/1D} agonist to treat social communication deficit and/or irritability in autism spectrum disorder (ASD). ML-004 is a reformulation of zolmitriptan (acute migraine drug), and while this is a much higher risk program, there are currently no FDA-approved therapies for ASD, and it would offer upside to the stock if successful. As such, we ascribe small, highly risk adjusted credit for this program currently. Elsewhere, MPLT has two other programs in treatment of motor symptoms of Parkinson's (ML-021) and hyperactivity (ML-009) respectively, though they are yet to advance to IND-enabling studies.

MapLight Pipeline

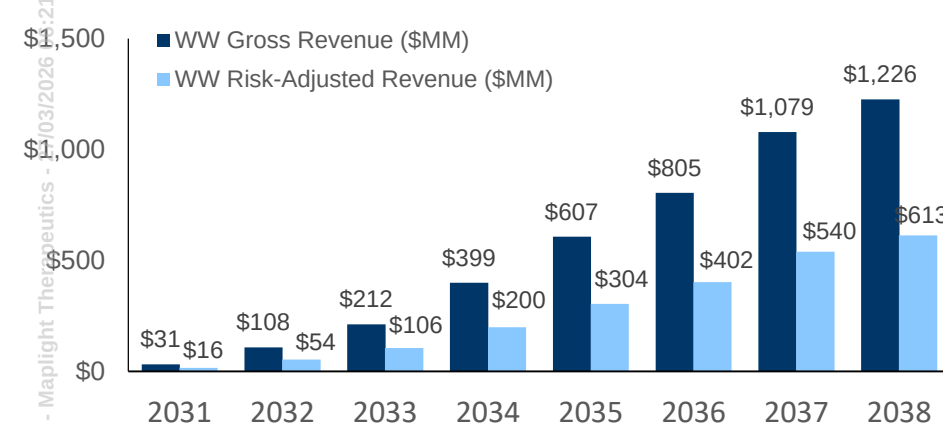
Program	Circuit	Indications	Preclinical	Phase 1	Phase 2	Phase 3	Anticipated Milestones
ML-007C-MA M/M agonist co-formulated with PAC	Direct and Indirect Pathways	Schizophrenia	ZEPHYR				Topline results in 2H 2026
		Alzheimer's Disease Psychosis	VISTA				Topline results in 2H 2027
ML-004 5-HT _{1B/1D} agonist	Dorsal Raphe to Nucleus Accumbens	Autism Spectrum Disorder Sociability/Irritability	IRIS				Topline results in 2H 2026
ML-021 M ₄ antagonist	Direct Pathway	Parkinson's Disease					IND-enabling studies complete in 2H 2026
ML-009 GPR52 PAM	Indirect Pathway	Hyperactivity/Impulsivity					Preclinical candidate nomination in 2026

MPLT's ph2 global trial in ASD is currently ongoing, and is expected to randomize 150 subjects with ASD, assessing change in baseline at 12wks in ABI, social communication domain score. We currently ascribe small, highly-risk adjusted credit for this program in our model, which represents upside along with MPLT's earlier-stage programs in Parkinson's disease (ML-021) and hyperactivity/impulsivity (ML-009).

SCZ and ADP Represent a Large Opportunity for MPLT: Even with Conservative Penetrations, ~\$3B Gross Opportunity Across The Two in 2038

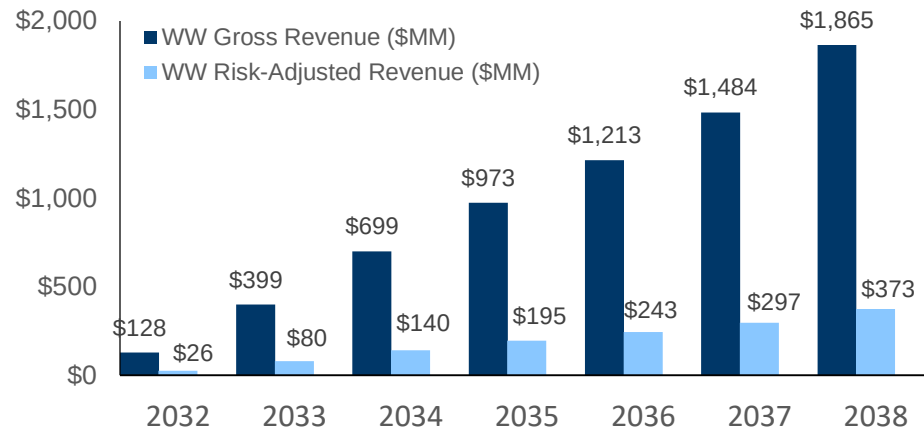
Based on KOL feedback, we estimate that the muscarinic class will capture up to 30% of treatment-resistant schizophrenia patients by 2038. We still see it as an open question how many muscarinic drugs make it to market – but within the class we assume MPLT can capture meaningful minority share. Meanwhile for ADP there's less competition at the moment, but it'll depend on if/when Cobenfy may receive approval (even if ADEPT-2 fails, there are two other shots on goal, including a RWT that probably still works). For now, we're modeling 20% penetration if successful, and credit here is highly risk adjusted. We also modeled minor contributions from ML-004 in ASD that are heavily risk-adjusted with 5% probability of success.

Schizophrenia Estimates



We assume that muscarinic antipsychotics will garner meaningful uptake for treatment-resistant schizophrenia patients. And among those muscarinics, we model Cobenfy still taking the lion's share with its first-mover advantage but also left room for other M4 competitors (e.g., NBIX) to be conservative.

Alzheimer's Disease Psychosis Estimates



Although ADP is less prevalent than schizophrenia, there's greater unmet need with no therapies approved for this specific indication. We model up to 20% penetration in 2038.

We Currently Derive A \$28 Target Price For MPLT, ~80% Upside To The Stock's Recent Close

Despite only having ph1 HV data, our valuation assumes a 50% probability-of-success for ML-007C-MA in schizophrenia due to sharing a similar mechanism with Cobenfy, while 20% probability-of-success in ADP is lower as we've yet to see pivotal data for Cobenfy in this indication -- that said, we believe MPLT has a potential shot at success even if Cobenfy's ADEPT-2 data falls short.

MPLT DCF - \$MM	2025E	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E	2038E	TV
CFO	(116)	(121)	(117)	(141)	(159)	(177)	(140)	(112)	7	135	297	399	484	532	
CFI	55	0	0	0	0	0	0	0	0	0	0	0	0	0	
Net Borrowing	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
FCF	(61)	(120)	(117)	(141)	(159)	(177)	(140)	(111)	8	135	297	399	484	532	4,835
Period	0.00	1.00	2.00	3.00	4.00	5.00	6.00	7.00	8.00	9.00	10.00	11.00	12.00	13.00	
PV FCF	(61)	(108)	(95)	(103)	(105)	(105)	(75)	(54)	3	53	105	127	138	137	1,245
Total FCF, PV	1,102.5														
Net Cash 4Q25E (Including IPO)	462.0														
Implied Market Capitalization	1,564.5														
Diluted Shares Outstanding 4Q25E (Including IPO)	55.9														
NPV Per Share	\$28														
Discount Rate	11.0%														
Terminal Growth Rate	0.0%														

Source: Company Filings, Stifel estimates. Note: Diluted share count includes options and RSUs.

MPLT P&L: We Model Cash Post IPO Being Sufficient To Fund Through Schizophrenia and ADP Ph2 Readouts

MapLight Therapeutics (MPLT) - IS Data in \$MM, Except EPS	2023	2024	1H25	3Q25E	4Q25E	2025E	2026E	2027E
Product Revenue	-	-	-	-	-	-	-	-
Total revenue	-	-	-	-	-	-	-	-
COGS	-	-	-	-	-	-	-	-
R&D	49.7	68.5	46.6	25.0	26.0	97.6	110.0	100.0
SG&A	7.6	14.4	7.6	4.0	4.2	15.8	20.0	30.0
Other operating expenses	-	-	-	-	-	-	-	-
Operating expenses	57.3	82.9	54.2	29.0	30.2	113.4	130.0	130.0
Operating income	(57.3)	(82.9)	(54.2)	(29.0)	(30.2)	(113.4)	(130.0)	(130.0)
Interest income	1.1	4.5	1.5	1.0	1.0	3.5	-	-
Interest expense	-	-	-	-	-	-	-	-
Other income (expense)	0.0	1.8	0.5	-	-	-	-	-
Net interest and other income	1.6	5.4	2.0	1.0	1.0	3.5	-	-
EBIT	(55.7)	(81.1)	(52.2)	(29.0)	(30.2)	(113.4)	(130.0)	(130.0)
Net pre-tax income	(55.7)	(77.6)	(52.2)	(28.0)	(29.2)	(109.9)	(130.0)	(130.0)
Provision for income taxes	0.5	-	-	-	-	-	-	-
Net income (loss)	(56.2)	(77.6)	(52.2)	(28.0)	(29.2)	(109.9)	(130.0)	(130.0)
Diluted earnings per share (GAAP)	\$ (5.07)	\$ (6.27)	\$ (4.07)	\$ (2.14)	\$ (0.67)	\$ (5.34)	\$ (2.61)	\$ (2.55)
Basic earnings per share (GAAP)	\$ (5.07)	\$ (6.27)	\$ (4.07)	\$ (2.14)	\$ (0.67)	\$ (5.34)	\$ (2.61)	\$ (2.55)
Average shares outstanding, basic	11.1	12.4	12.8	13.1	43.6	20.6	49.9	50.9
Average shares outstanding, diluted	11.1	12.4	12.8	13.1	43.6	20.6	49.9	50.9
Source: Stifel estimates & company filings								
MapLight Therapeutics BS/CFS (\$MM)	2023	2024	1H25	3Q25E	4Q25E	2025E	2026E	2027E
Net cash	158.9	65.3	60.5	221.0	462.0	462.0	591.6	475.1
Cash and equivalents	106.8	65.3	60.5	221.0	462.0	462.0	591.6	475.1
Debt	-	-	-	-	-	-	-	-

Source: MapLight Materials; Stifel Estimates.

MapLight Therapeutics: Risks To Investment

- **Clinical Risk:** MapLight's lead programs are focused in neuropsychiatry, an area of research that is highly challenging for drug development. ML-007C-MA is not an FDA approved product yet and future trials carry risk. And while there is prior data for Cobenfy, a drug with a similar mechanism approved for schizophrenia, Cobenfy has not yet been approved for Alzheimer's disease psychosis and this indication is arguably more challenging.
- **Competitive Risks:** MapLight competes with ABBV, BMY, NBIX, NMRA, LBRX, among others, who are currently marketing or in pursuit of novel treatments in similar indications. It is no guarantee that if MapLight is successful clinically that they will achieve commercial success. Moreover, MPLT shares may move on other company's data readouts (i.e. ADP) which presents a risk if data are negative.
- **Regulatory Risk:** All of MapLight's programs face regulatory risk related to the need for approval by the FDA or other global regulatory agencies before commercial launches.
- **IP Risk:** MapLight's investors face intellectual property risk related to the company's own patents and competitors' patents.
- **Financing Risk:** MapLight is likely to pursue additional equity financings in the future that may be dilutive to the value of publicly traded shares.

Delving Further Into ML-007C-MA And Our Diligence on MPLT

MapLight Is Currently Running Ph2 Studies in Schizophrenia, ADP and ASD; Investor Focus Is On The Muscarinic Clinical Programs

MPLT has two clinical-stage assets: ML-007C-MA, an M1/M4 agonist which is in ph2 development for both schizophrenia and Alzheimer's disease psychosis, and ML-004, a 5-HT_{1B/1D} agonist also in ph2 for social communication deficits and/or irritability associated with Autism Spectrum Disorder. Most of the interest is currently focused on schizophrenia and ADP where the M1/M4 mechanism has been validated or is currently being validated in pivotal studies.

Program	Circuit	Indications	Preclinical	Phase 1	Phase 2	Phase 3	Anticipated Milestones
ML-007C-MA M/M ₄ agonist co-formulated with PAC	Direct and Indirect Pathways	Schizophrenia	ZEPHYR				Topline results in 2H 2026
		Alzheimer's Disease Psychosis	VISTA				Topline results in 2H 2027
ML-004 5-HT _{1B/1D} agonist	Dorsal Raphe to Nucleus Accumbens	Autism Spectrum Disorder Sociability/Irritability	IRIS				Topline results in 2H 2026
ML-021 M ₄ antagonist	Direct Pathway	Parkinson's Disease					IND-enabling studies complete in 2H 2026
ML-009 GPR52 PAM	Indirect Pathway	Hyperactivity/Impulsivity					Preclinical candidate nomination in 2026

Source: MapLight Materials; Stifel Research.

Cobenfy's Success Has Paved The Way For Followers Hoping To Enter The Blockbuster Muscarinic Market, But Some Encountered Setbacks

As the muscarinic space has become exciting, MPLT is not the only player as other companies have also tried to follow in KRTX's footsteps. Each is trying to push the envelope on mechanism/drug design in hopes of showing incremental benefits and differentiation from Cobenfy, but some approaches, notably emraclidine, have not panned out as expected.

Company	Ticker	Drug (Mechanism)	Development Phase	Catalysts	Notes
 Bristol Myers Squibb	BMY	Cobenfy (M1/M4 agonist)	FDA approved	—	In pivotal studies for ADP
 MapLight	MPLT	ML-007C-MA (M1/M4 agonist)	Ph2	Ph2 topline in 2H26	Also in ph2 for ADP
 abbvie	ABBV	Emraclidine (M4 PAM)	Ph2	—	Both ph2 trials failed to meet primary endpt
 NEUROCRINE BIOSCIENCES	NBIX	NBI-'568 (M4 agonist) NBI-'570 (M1/M4 agonist) NBI-'567 (M1-preferring dual agonist) NBI-'569 (M4-preferring dual agonist)	'568 in ph3 '570 in ph2 '567, '569 in ph1	'567 and '569 topline in late 2025	'568 inverse dose response
 Neumora™	NMRA	NMRA-861 (M4 PAM) NMRA-898 (M4 PAM)	'861 in ph1 '898 in preclinical	M4 franchise update in mid-2026	

Source: Company Materials; Stifel Research.

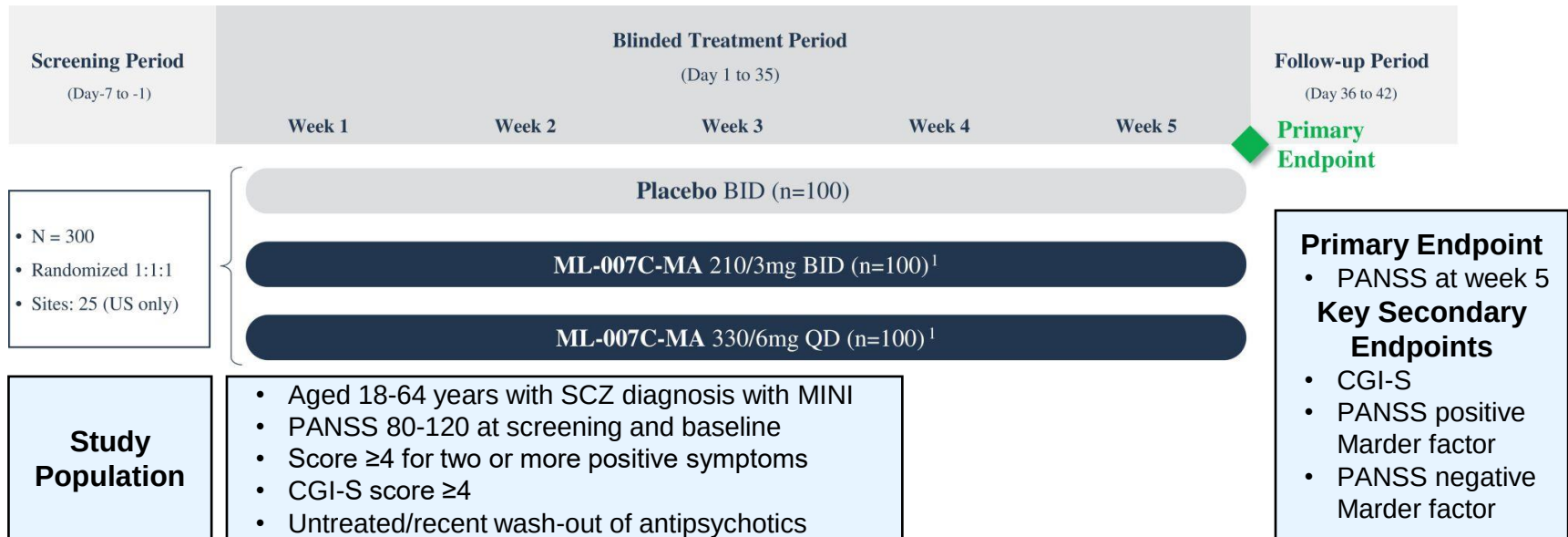
MPLT Leverages the Learnings from KRTX, CERE, and NBIX – By Design, ML-007C-MA Is On the Right Side of the Various “Ideal Muscarinic” Debates

As the muscarinic space has become exciting, multiple other companies have tried to follow in KRTX’s footsteps but have encountered setbacks. For example, although the exact reasons for ABBV/CERE’s emraclidine’s failure in ph2/3 are unknown (probably multiple factors), we think it’s reassuring that MPLT’s drug is more similar to Cobenfy than emraclidine: (1) the importance of hitting M1 on top of M4 is unclear, but it’s less risky to just target both which is what ML-007 does, and (2) agonists like ML-007 can exert their effects independently, vs PAMs that need the natural muscarinic ligand. Meanwhile, MPLT is hoping to achieve more convenient dosing than Cobenfy which is BID in schizophrenia and TID in ADP. Based on ph1 PK modeling, they believe it’s possible that QD dosing will be sufficient, though MPLT is advancing both a QD and BID regimen in the SCZ ph2 (and BID in ADP) to be safe. Bottom line: none of these factors necessarily guarantee success, but we feel more confident extrapolating the Cobenfy data to MPLT’s molecule, than we would for say NBIX (selective M4 agonist) or ABBV/CERE (selective M4 PAM).

PAM 	<i>While the reason for emraclidine’s failure can be debated, from a biological perspective, PAMs require the natural ligand to be present near the receptor whereas agonists do not. So, agonists are more likely to be de-risked and efficacious under broader set of circumstances.</i>	Agonist 
M4 Selective 	<i>Perhaps a more widely debated topic is the importance of M1: some animal experiments show that M1 is necessary for antipsychotic efficacy while others suggest it’s incremental. But Cobenfy (M1/M4) is the only one that has succeeded in ph3 so far--pursuing a similar approach is less risky.</i>	M1/M4 Selective 
No Anticholinergic 	<i>Both KRTX and MPLT use a peripheral antagonist to minimize side effects, while the M4-only candidates do not. Emraclidine did show a better tolerability profile than Cobenfy, but they may have been underdosing and we haven’t seen details from the ph3.</i>	With Anticholinergic 
Twice Daily Dosing 	<i>Less frequent dosing is always better, especially in a neuropsychiatric disease where compliance is an issue. Pill burden also matters when a food effect needs to be navigated. Note that MPLT is aiming for once daily dosing in SCZ, though they are advancing both QD and BID dosing in ph2.</i>	Once Daily Dosing 

Schizophrenia Phase 2 Study Is Set to Have Topline Data 2H26; This Is Large At Least Partially De-risked By Cobenfy's Positive EMERGENT-1/2/3 Studies

The ZEPHYR Phase 2 Trial Design

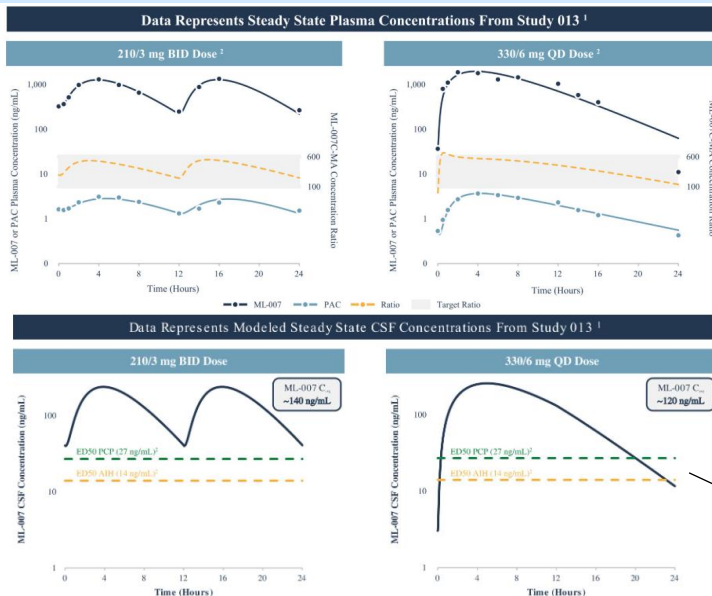


MapLight recently initiated a phase 2 study of their M1/M4 agonist ML-007C-MA in schizophrenia. This trial builds on the ph1 results in HVs and will explore both once and twice daily dosing. The design here is similar to Cobenfy's ph2 (EMERGENT-1), including the primary endpoint of change in PANSS from baseline to week 5. MapLight is aiming to enroll ~300 patients in the US (KRTX enrolled 182 patients across 2 arms), which could make this study big enough to count as a pivotal trial

Can MPLT Really Achieve QD? PK Curves Suggest Both QD and BID Achieve Good Exposures, And Safety Doesn't Seem Materially Different Either

Dosing frequency makes a meaningful difference especially in indications such as schizophrenia and ADP where psychosis and impaired mental status hinders compliance. MPLT is hoping for QD dosing for schizophrenia but has also hedged their bet by advancing both a BID and QD dose arms into ph2. On efficacy, ML-007 plasma concentrations for both BID and QD seem to suggest similar exposures, though peak-to-trough for QD is understandably larger. Modeled CSF concentrations for QD do briefly fall below the 14-27ng/mL threshold established in mouse models, though the average concentration is much higher. Meanwhile, the slightly higher peak with QD hasn't translated to obviously worse safety/tolerability compared to BID in HV. Bottom line: there's a solid chance MPLT will be able to advance QD dosing without issues, which we think could be a significant differentiator.

QD vs BID Plasma and CSF Concentrations



QD vs BID Safety and Tolerability

	Placebo Combined (N = 8) ²	210/3 mg BID (N = 6)	330/6 mg QD (N = 6) ³
Subjects with Any TEAE ¹	2 (25%)	3 (50%)	4 (67%)
Mild	2 (25%)	3 (50%)	4 (67%)
Moderate ³	0 (0%)	1 (17%)	2 (33%)
Severe	0	0	0
Most Common TEAEs (>1 Subject in any Dose Group)			
Chills	0	0	2 (33%)
Hyperhidrosis	0	1 (17%)	2 (33%)
Nausea	0	3 (50%)	3 (50%)
Dizziness	0	3 (50%)	2 (33%)
Dyspepsia	0	0	2 (33%)

The 14-27ng/mL ED50 values in CSF were determined using mouse models. QD did exceed this for majority of the time course. The difference in exposure between BID and QD probably won't impact efficacy.

There were slightly more TEAEs in the QD group (chills, dyspepsia), but the sample sizes are limited, and nausea rates remain similar (and no vomiting).

On Safety, MPLT Has Rigorously Optimized The Ratio Of ML-007 vs PAC Which Should Hopefully Translate To Improved Tolerability

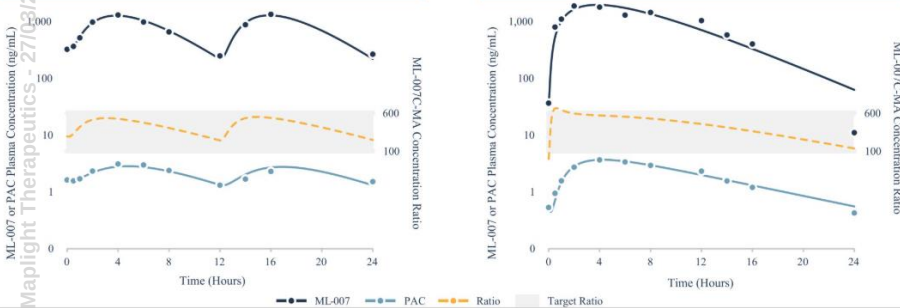
MPLT believes that ML-007:PAC plasma exposure ratios between 100:1 to 600:1 are the ideal range, determined empirically from their ph1 study (011) using measures of cholinergic activity (salivary volume changes) and frequency of adverse events, both procholinergic (nausea, vomiting, diarrhea, hyperhidrosis) and anticholinergic (dry mouth, tachycardia, dry eye). While this is a fair approach, the underlying assumptions for MPLT's drug to be more tolerable than Cobenfy are that (1) the ideal pro- to anti-cholinergic plasma ratios are translatable between HV and schizophrenia patients, and (2) Cobenfy does NOT fall within its ideal range. For context, Karuna (at PureTech originally) also did a ton of work on dose optimization, but they may have been constrained by the challenging med-chem profile of xanomeline, which is great drug but also a very old drug. Our view is that MPLT has done good work here and that there is room to improve over KarXT/Cobenfy on AEs.

"Ideal" Ratio between Pro- and Anti-Cholinergics

Data Represents Steady State Plasma Concentrations From Study 013¹

210/3 mg BID Dose²

330/6 mg QD Dose²



Did KRTX Test Different Ratios For SCZ?

Study	Xanomeline Dose	Trospium Dose
Ph1 POC	75mg TID	-
	75mg TID	20mg BID
Ph1 Co-formulation	100mg BID	20mg BID
	125mg BID	30mg BID
	150mg BID	40mg BID
Ph2 and Ph3 studies	50mg to 125mg BID	20mg to 30mg BID

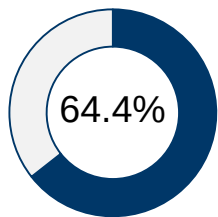
From study 011, MPLT established the ideal plasma ratio between their two compounds, which their co-formulated tablet was able to conform to in study 013. Though a key question is whether this ratio of optimal tolerability will translate from HV to schizophrenia patients.

In the ph1 Cobenfy studies, only a few combinations of xanomeline and trospium doses were explored before the final 50mg/20mg, 100mg/20mg, and 125mg/30mg dose titration regimen was selected for ph2 and ph3 studies. Further optimization could be possible in theory.

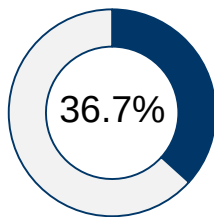
Further Confirmation That There's Room to Differentiate: Procholinergic AEs Are Prevalent In Real-World, And Some Docs Resort To Increasing Trosipium

A recent survey collecting real-world data on Cobenfy showed greater rates of GI adverse events than in clinical studies. Nausea and vomiting, both procholinergic signals, were commonly reported, and the survey did not call out any anticholinergic AEs. The natural conclusion is that the xanomeline to trosipium ratio in the standard KarXT doses is skewed towards procholinergic xanomeline, and docs have been increasing trosipium doses to compensate. Bottom-line: While worse tolerability from clinical trials to real-world is expected to some degree, it does show that there is plenty of room for MPLT to improve on tolerability.

Higher Frequency Of GI Adverse Events In Real-World Survey (n=90)



Nausea

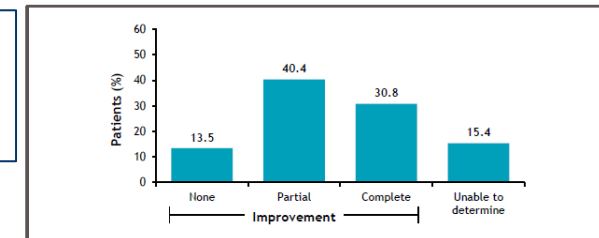


Vomiting

- Among patients who experienced nausea or vomiting
 - 55% received pharmacologic intervention
 - 21% had their dose of X/T reduced
 - 24% required no intervention
- The most common pharmacologic interventions were ondansetron and additional trosipium

Mitigation Strategies for GI Adverse Events

Figure 5. Effect of pharmacologic treatment on nausea and vomiting



About 2/3 and 1/3 of patients captured by BMJ's survey reported nausea and vomiting, respectively, after initiating Cobenfy. This is an interesting contrast against the 19% and 15% observed in the two ph3 studies, especially since (1) docs were already titrating slowly in the real-world and (2) prophylactic antiemetics were already prescribed in 21.1% of patients

To mitigate the GI side effects, about half of patients received pharmacological intervention, most commonly either the antiemetic ondansetron or additional trosipium dose, and a quarter of patients had their KarXT dose reduced. Additional trosipium makes sense since its anticholinergic effects should cancel out procholinergic AEs like nausea and vomiting, but at the same time implies that the original KarXT doses are not balanced

Heart Rate And BP Changes Have Been Observed With Muscarinics, And We Expect MPLT To Have Similar Signals

3-4 years ago, there was a lot of debate among investors on heart rate and blood pressure increases seen with Cobenfy and emraclidine and whether there would be any regulatory implications. Karuna later ran an ambulatory blood pressure monitoring study per FDA recommendations, which showed that the BP changes observed in trials was really just a small Cmax effect, whereas HR increases were sustained but in the grand scheme fairly small. FDA is far more concerned about BP, so with that cleared, the final Cobenfy label only recommended heart rate measurements as needed. Since ML-007C-MA also targets M1/M4, our expectation is that we may also see small changes in BP at Cmax but we expect a future ABPM study to show an overall benign profile here. From the ph1 ML-007C-MA results so far, the heart rate increases are transient, dose-dependent, and smaller in elderly vs healthy adults, all of which are reassuring.

FDA Is Wary Towards Drugs That Inadvertently Increase BP Due To CV Risk

"There is greater concern with the effect of a drug on blood pressure when the drug will be used chronically. As noted above, epidemiologic studies show that risk is related to blood pressure as a continuous function, and that sustained increases in blood pressure correlate with long-term increased risk of cardiovascular adverse events. It follows that even small, sustained increases in blood pressure (2 to 3 mm Hg) chronically would be expected to have such an effect. Thus, detecting such changes is important for drugs intended for chronic use, and for this reason, a sponsor should include a thorough blood pressure assessment, as described in this guidance, for a drug intended for chronic use. **As discussed in section IV. Considerations for Ambulatory Blood Pressure Monitoring, FDA recommends use of ABPM for this assessment, as ABPM is capable of detecting small, but potentially relevant, blood pressure effects.** ABPM also assesses effects over a 24-hour period, more relevant than a single time point"

– FDA (2018), Emphasis added by Stifel

Cobenfy Ultimately Didn't Increase Ambulatory BP, But Heart Rate Increase Still On Label

KRTX ran a ph1b open-label 8wk study assessing 24-hr ambulatory BP in adults with schizophrenia that showed a reassuring decrease of -0.59mmHg, but at the same time, there was also a mean increase in 24-hr heart rate by 9.8bpm at week 8 vs BL. As such, the Cobenfy label doesn't explicitly call out blood pressure risks, though it does warn providers to measure heart rate before and after initiating drug.

"Increases in heart rate: Cobenfy may increase heart rate. Assess heart rate at baseline and as clinically indicated during treatment with Cobenfy." - FDA label for Cobenfy (2025)

A Few Cases of Tremor Were Reported in MPLT's Healthy Elderly Study – Is This of Concern?

Of note, MPLT saw tremor in 27% (n=3/11) of elderly participants, which exceeds rates observed with Cobenfy (0.4%). For context, tremor can occur with muscarinic agonists, though the prevailing hypothesis is that this is through the M2 pathway. MPLT describes ML-007's affinity to M1/M4 vs other muscarinic receptors as similar to Cobenfy, but due to higher CNS exposures, it's entirely possible that this could be real and on target. Notably though, these tremors were described to have minimal impact on patients' function, and MPLT's 13-week chronic dog tox study also confirms these abnormal movements are not seizure-related, which is reassuring (seizure was seen in prior preclinical work with NMRA's compound). Bottom line: given that this was a small study and the events were mild, we don't know yet if this is a real clinically significant finding, or if it's just noise and something that will largely be mitigated by titration, but at face this isn't overly concerning.

Older Muscarinics Are Tremorogenic, But Xanomeline Is Not?

Muscarinic	Effect	Source
pirenzepate	Induced tremor in rats	Collins 2010
oxotremorine	Induced tremor in mice	Gomez 1999

Some muscarinic drugs--pirenzepate which preferentially targets M1/M2/M3 and oxotremorine, a non-selective muscarinic agonist--induce tremors in animal models, and ML-007 has shown similar tremorogenic effects in healthy elderly (27%) and animal studies. Cobenfy only saw 0.4% tremor rates in long-term studies

M2 Thought to Play a Role in Tremors



Genotype	Tremor with Oxotremorine?
M2 ^{+/+}	Yes
M2 ^{+/-}	No
M2 ^{-/-}	No

Tremors are thought to be from M2 agonism in the brain based on partial knockout and knockout mice studies. Mgmt described ML-007's affinity to M1/M4 vs other muscarinic receptors as similar to xanomeline, though with higher CNS exposures, we're not surprised to see some M2 signals

No EEG Abnormalities and Tremors Self-Resolved



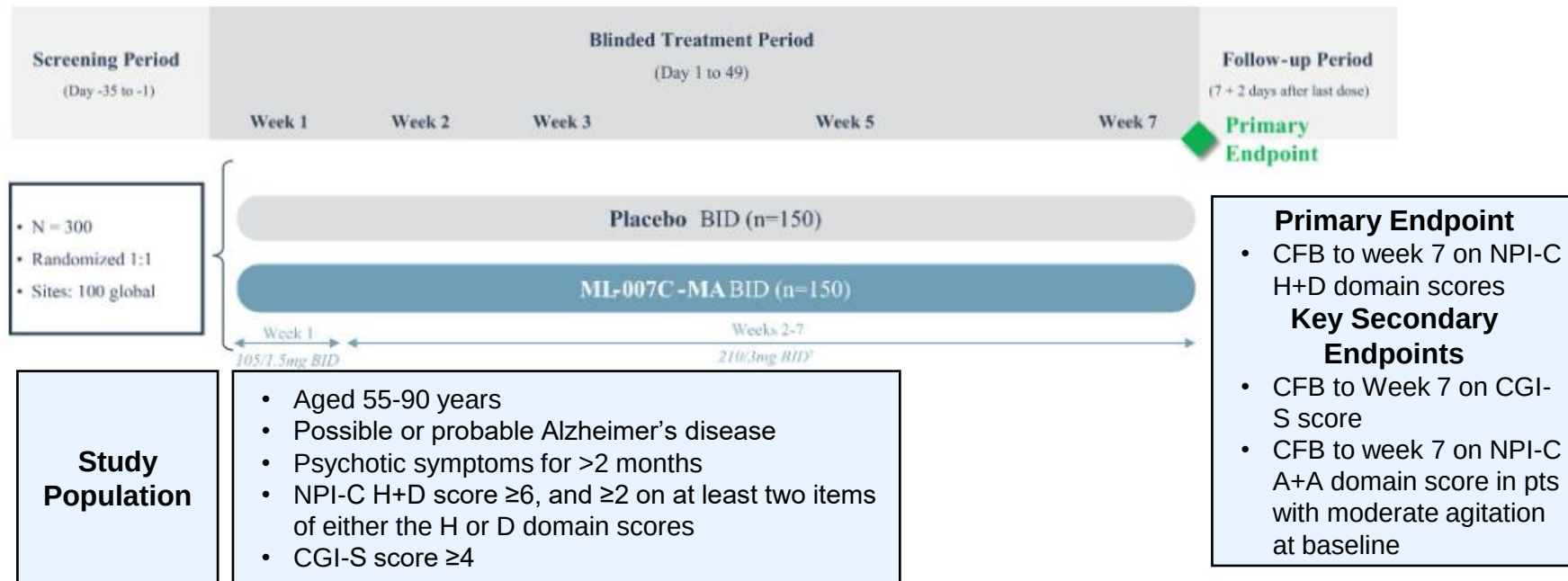
- 13-week ML-007 dog tox study conducted
- No EEG abnormalities even during tremors, suggesting that these are not seizure-related



- Tremors observed with ML-007C-MA in 3 of the 11 elderly patients in their ph1 study
- Tremors did not interfere with function or ADLs

For ADP, The Phase 2 VISTA Study Will Also Be First Proof-Of-Concept Data For Their Drug In This Indication--Readout Expected In 2H27

The VISTA Phase 2 Trial Design



In parallel to the ZEPHYR study, MPLT is also running ph2 VISTA in ADP. This will be a global trial with n=300 participants randomized 1:1 between placebo or 210mg/3mg BID doses of MIL-007C-MA. The study design is similar to ADEPT-2, including a primary endpoint of NPI-C:H+D. The trial also has similar inc/exc criteria, though BMJ is looking at a wk 14 primary endpt while MPLT is assessing wk 7.

Cobenfy's ADEPT-1, ADEPT-2, and ADEPT-4 Studies May Provide Confirmation That Muscarinics Are Effective In ADP, But Trial is Not Without Risk

Unlike schizophrenia where Cobenfy has already succeeded in pivotal studies, there's relatively less evidence de-risking the ADP opportunity for MPLT. For context, xanomeline, the cholinergic component of Cobenfy, was originally developed by Lilly for Alzheimer's where it demonstrated efficacy on cognitive scales in a ph2 study, an important finding that validated exploration of muscarinics in this indication. Moreover, there was a surprise finding in a post-hoc analysis, where LLY observed a dose dependent beneficial impact on the frequency of both hallucinations and delusions. Cobenfy's ph3 pivotal program will be the first to verify this benefit in psychotic symptoms in a prespecified analysis. ADEPT-2 will be the first to readout later this year, followed shortly by ADEPT-1 and ADEPT-4 in 2026. BMY only needs 2 of the 3 pivotal studies to be positive to hopefully secure FDA approval in ADP. As it stands though investor expectations here are low, because BMY's public commentary hasn't been all that encouraging.

Older Xanomeline Studies Showed Signals

	Dose (mg)	ADAS-Cog		CIBIC+	
		Completers	Endpoint	Completers	Endpoint
No. of pts	225	38	78	38	79
	150	45	75	46	75
	75	68	83	50	83
	0	58	83	57	85
Mean diff	225	-1.42	0.04	4.00	4.16
	150	0.38	1.34	4.09	4.11
	75	1.03	1.06	4.44	4.41
	0	1.42	1.35	4.33	4.34
Adjusted mean	225	20.96	22.41	3.93	4.13
	150	22.46	23.72	4.01	4.09
	75	23.20	23.37	4.40	4.41
	0	23.80	23.67	4.30	4.32
Dose-response P		.033	.342	.005	.014
P vs pbo	225	.045	.221	.022	.014
	150	.367	.557	.036	.057
	75	.935	.955	.846	.017

* Completers were those reaching 6 months of therapy; endpoint refers to last measurement collected during double-blind period

High-dose xanomeline demonstrated statistically significant improvement vs. PBO on cognitive scales (ADAS-Cog, CIBIC+) in a large ph2 study in Alzheimer's (i.e., "probably AD"). There was also a compelling decrease in the rate of hallucinations and delusions. The discontinuation rate for the high dose though was 50%, largely due to nausea/vomiting, while LLY also saw an imbalance in syncope.

KarXT Will Retest ADP Hypothesis With Ph3 Studies

Study Design	- Ph.3, pivotal, randomized, placebo-controlled, outpatient study 380~400 patients in each study are randomized to receive KarXT or placebo.
Inclusion/Exclusion	- Patients age 55-90 - Possible or probable Alzheimer's Disease - Caregiver willing to report subject status and check compliance with medication and study procedures - Score ≥ 2 on NPI-C delusions or hallucinations domain
Primary Endpoint	- ADEPT-1: Time from randomization to relapse within 38-week study - ADEPT-2/4: Change from BL to week 14 on the NPI-C:H+D score
Dosing	T1D with 20/2mg, 30/3mg, 40/4mg, 50/5mg and 66.7/6.67mg doses available for titration

Investor Expectations for Cobenfy In the First Ph3 ADP Readout (ADEPT-2) Are Low; To Understand MPLT Implications We Will Need the Full Data

Naturally, we expect investors to be paying close attention to the outcome of Cobenfy's ADP program, which is currently in phase 3 and expecting the first of multiple readouts by the end of this year (ADEPT-2). Generally speaking, expectations are lower on this first ph3 readout: BMJ has referred to conducting clinical trial site reviews prior to readout (suggesting that there may have been a conduct issue) and further, the delays to the readout coupled with the higher-risk profile of the program has driven some skepticism here. While we'd view success here (and for the program more broadly) as a clear win for MPLT that should send the stock a lot higher, a failure could be negative but it also could be nuanced – for example a clear efficacy trend but a study that failed due to compliance issues, dropouts or patients not being able to titrate all the way up to the highest dose would arguably lay out a potential path to success for MPLT

Challenges with Execution

We've seen examples of a company getting acquired and subsequently fail late-stage studies—most notably, competitor muscarinic company CERE (acquired by ABBV) failed in its pivotal schizophrenia studies. Changes in management can lead to oversight on execution, which is especially important in psychiatry studies where patient selection and managing placebo response are so critical. BMJ has alluded to this potentially being a factor, noting they're conducting a trial site review prior to disclosing the results

"Specifically as it relates to ADEPT-2, in preparation of that database lock, we are conducting the clinical trial site reviews that could impact the timelines a bit." -- BMJ CMO, July 2025)

Poorer Tolerability in Elderly Patients

One of the key challenges of this MOA / drug target is challenges associated with tolerability, and we believe this is even more pronounced in an elderly population with Alzheimer's, who are particularly sensitive to side effects. This has been reflected in both BMJ/KRTX and MPLT conducting extensive ph1 work here with healthy elderly to inform dosing, which for BMJ/KRTX was TID dosing with a much slower titration. Also, we saw more detailed healthy elderly for KarXT in the FDA review docs, and AE rates were much higher. And while we think it's still early days for MPLT showing differentiation on tolerability in schizophrenia, in ADP, because MPLT is BID (not TID) and lacks a food issue, one could argue that a KarXT failure (depending on details) shouldn't imply that ML-007 will necessarily fail too.

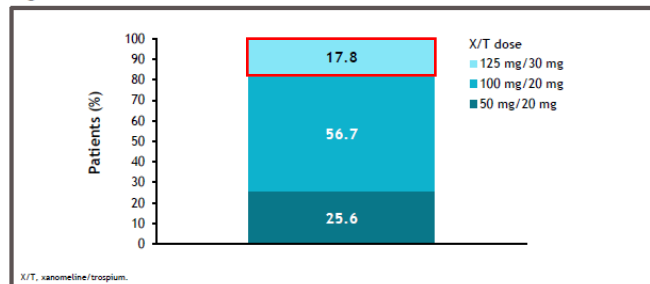
Bottom line: we'll have to see what the data show, but as ADP is an area where the ML-007 profile might be relatively more differentiated, the ADEPT-2 data will need to be interpreted in context

What Would Give Us Confidence On MPLT If ADEPT-2 Fails? An Efficacy Signal in Patients Able To Tolerate Highest Dose

Between the conclusion of the ADEPT-2 study (July 31, 2025 based on clinicaltrials.gov) and the YE readout, there's plenty of opportunity for BMJ to closely interrogate the data and present different subgroup analyses, in a scenario where the trial were to miss its primary outcome. Our expectation is that not all patients will be able to hit the max 66.7/6.67mg TID dose in ADP since tolerability was an issue in schizophrenia as well. But if their post-hoc analyses revealed greater improvements on the NPI-C:H+D endpoint in patients who were able to reach and maintain a higher dose of KarXT than those who didn't, that's a positive for MPLT, assuming the latter does turn out to be more tolerable in the elderly. Admittedly it remains TBD though on what BMJ will share in the top-line readout, and what commentary management will provide.

Schizophrenia Patients Aren't Able To Hit Max KarXT Dose, Even With Slow Titration

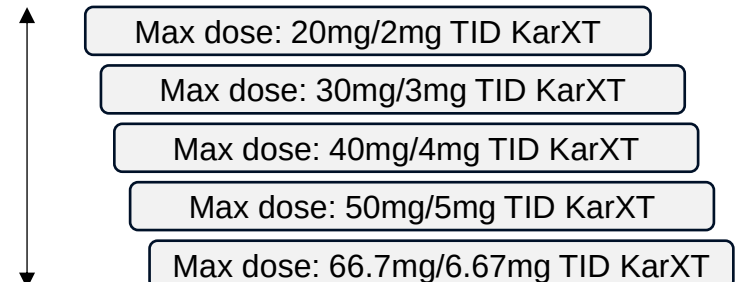
Figure 4. Maximum dose of X/T administered



BMJ's survey confirms that not all patients are able to reach or stay on max dose of KarXT in the real-world since GI tolerability tends to be an issue. In fact, among the n=90 patients, only <20% were at the max dose when the data were collected. To be fair, most surveyed patients (58%) were only KarXT for ≤3mo at the time, but 29% had already been on it for >6mo.

Subgroup Analysis on "High-Dose" Patients Will Be Most Important For MPLT

Minimal Readthrough to MPLT



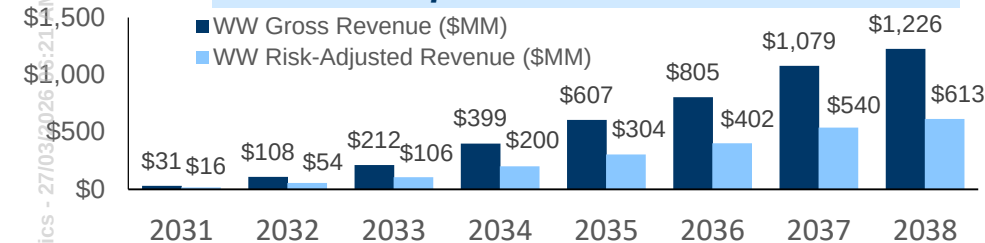
More Readthrough to MPLT

If ADEPT-2 patients that were able to tolerate a higher dose of KarXT showed greater improvement on NPI-C:H+D, that would validate the potential efficacy of muscarinics in this indication.

Commercial Opportunity In Schizophrenia: Even With Conservative Assumptions, We Model Blockbuster Sales By 2038 With Levers For Upside

In our view, schizophrenia alone already represents a blockbuster potential opportunity for MPLT's drug. At a future steady-state, we're estimating roughly 30% of treatment-resistant schizophrenia patients could be on muscarinics based on recent doc feedback. We also model a 2031 schizophrenia launch for MPLT that could capture ~20% of share by 2038. Even with these conservative penetration estimates, ML-007C-MA could still achieve blockbuster sales, with room for further upside if (1) there is eventually positive data for muscarinics as an adjunctive treatment, (2) M4-only approaches fall short and lessen the competition, and (3) MPLT does show clearly better efficacy and/or safety than Cobenfy. We model a 50% probability of success for MPLT in schizophrenia.

Schizophrenia Sales Estimates



1.3MM tx-resistant SCZ in US

X

30% muscarinic penetration

X

20% ML-007C-MA share within muscarinics

X

~\$27,000 gross annual cost, 40% GTN, 60% compliance

=

~\$1.1B US revenue in 2038

1.6MM non tx-resistant

X

3%

X

Points To Consider



- KarXT failed in adjunctive study; only a minority of KOLs currently use it as adjunct to dopaminergic antipsychotics
- Future success in adjunctive study could sharply increase muscarinic utilization



- While NBIX and its ph3 for '568 is currently ahead of MPLT, the ph2 data doesn't perfectly de-risk the M4-only approach
- If MPLT is 2nd to market, it could achieve higher share within the muscarinic class

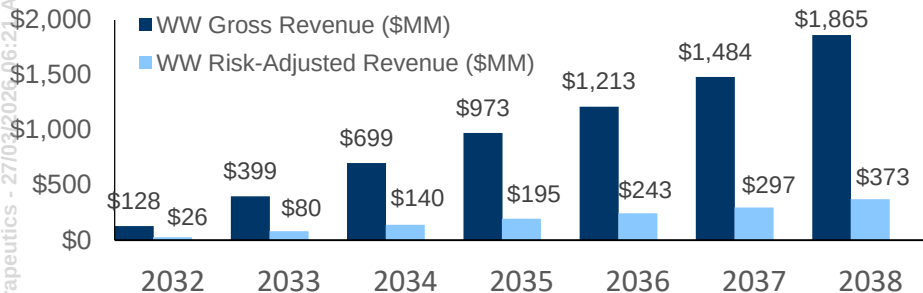


- Relative market share is also dependent on the eventual target product profile
- We're optimistic on MPLT's ability to achieve QD dosing, but if it also shows clear efficacy and/or tolerability benefits, that's a plus

Commercial Opportunity In ADP: High-Risk / High-Reward With Potential For ML-007C-MA To Be First Approved Therapy In This Indication

At this point, it's hard to really model ADP without knowing the data or the actual potential product profile. As we see it, a new ADP drug could sell ~\$1B or ~\$5B – it really depends on the magnitude of the efficacy signal, and, potentially more importantly, tolerability and safety. Thus, for now we model a ~\$2B opportunity which is based on a conservative penetration assumption, in the absence of actual clinical data. We outline our thoughts below on how we think about the model here.

Alzheimer's Disease Psychosis Sales Estimates



~800,000 diagnosed ADP patients in US

X

20% ML-007C-MA penetration

X

~\$27,000 gross annual cost, 40% GTN, 60% compliance

=

~\$1.6B US revenue in 2038

Points To Consider



- With no approved drugs specifically for ADP, the market is nascent relative to SCZ and may require more physician education
- Though high caregiver burden in this disease is a strong motivation to seek treatment



- Even if ADEPT-2 fails, ADEPT-1 and ADEPT-4 should be sufficient for KarXT approval, if both are positive
- MPLT being 1st vs 2nd to market will have meaningful implications to its market share



- There's more room to differentiate in ADP where KarXT's TID dosing with fasting may be a struggle in these elderly frail patients
- Aside from BID dosing, we'll also have to see if tolerability with MPLT ends up superior

MPLT Continues to Strengthen their Patent Portfolio and are Expecting Exclusivity Into Mid-2040s If Not Longer

While IP for ML-007 itself does not extend far out, MPLT is currently in the process of filing patents for the use of ML-007 in conjunction with a peripheral anticholinergic agent for the treatment of schizophrenia and ADP. They anticipate these patents, once approved, could extend exclusivity into mid-2040s, even before taking PTA/PTE into account (though we note that PTE may be used on the COM which would extend the floor into the mid/late 2030s). And while method patents are often discounted or harder to know if they'll hold up, we do note that these were the core IP for KarXT, underlying BMY's acquisition.

Patent Type	Projected Expiration Year	Status
Composition of matter for ML-007	2031-2032	Approved
Method of use (ML-007 in schizophrenia and ADP)	2031	Approved
Method of use (ML-007 + PAC in schizophrenia and ADP)	2042	Pending in US and 16 other foreign countries
Method of use (ML-007 + PAC in schizophrenia and ADP)	2044	Pending under the Patent Cooperation Treaty
Salts and crystalline polymorphs of ML-007	2045	Pending in US

- Projected expiration year does not take patent term adjustments, extensions, or other forms of exclusivity into account
- Fesoterodine COM patent has expired; no expected issues using this product in combination with ML-007

Lastly on ASD, ML-004 is an IR/ER Formulation of Zolmitriptan and MPLT Believes 5-HT_{1B/1D} Agonism Could Have Potential; Higher Risk, Little Credit in Stock Today

Zolmitriptan is an old drug approved for as-needed treatment of migraine headache. MPLT is developing a higher dose IR/ER oral tablet to evaluate its potential in addressing irritability and sociability issues associated with ASD. Based on optogenetic and mouse experiments, MPLT believes 5-HT_{1B} is intricately linked to sociability behavior in ASD models. In humans, MPLT conducted a ph1 HV evaluating safety/tolerability of high-dose IR zolmitriptan before another ph1 HV this time with their IR/ER formulation (ML-004). Both forms were generally well-tolerated. The ongoing ph2 expected to readout in 2H26 will be the first to evaluate efficacy in the ASD population.

Mechanistic Rationale for ML-004



- MDMA also targets 5-HT_{1B} receptor
- Mice treated with MDMA showed dose-dependent improvements in social behavior
- This was reversed by infusing a 5-HT_{1B} blocker into the nucleus accumbens



- Zolmitriptan was evaluated in mouse valproic acid model for autism, where prenatal VPA increases risk of ASD in offspring
- 10mg/kg of zolmitriptan significantly reversed social deficits in these mice



- CNTNAP2 knockout mice are also another model, since this gene is a risk factor for ASD and other neurodevelopmental disorders
- Treatment with 10mg/kg zolmitriptan also improved social deficit in knockout mice

Clinical Studies for ML-004

Study Design and Results

**Ph1
(N=40)**

- Randomized, DB, pbo-controlled MAD in HV
- 5mg, 10mg, 20mg, 30mg, 40mg TID oral IR zolmitriptan
- Mostly mild AEs, most commonly headache, dizziness, nausea and hiccups; higher frequency at 40mg dose

**Ph1
(N=12)**

- Open-label bioavailability and PK study in HV
- 24mg (6mg IR/18mg ER) zolmitriptan bilayer oral tablet
- Lower Cmax and longer plasma exposure
- Well-tolerated with AEs less common than IR

**Ph2
(N~150)**

- Ongoing ph2 in adult and adolescent ASD patients
- Flexible dosing paradigm
- Primary endpt is change in ABI-SC @ 12 weeks
- Well-tolerated with AEs less common than IR

MPLT Management Team (1/2)



Christopher Kroeger, MD, MBA
Chief Executive Officer and Founder

Dr. Kroeger brings more than 20 years of experience leading, building and advising development-stage therapeutic and medical device companies, combined with the expertise of a physician and scientist. Dr. Kroeger previously served as the CEO of Cardioxyl Pharmaceuticals, leading a team that produced three investigational new drug candidates and successfully executed six Phase 1 and Phase 2 clinical studies, culminating in the sale of the company to Bristol Myers Squibb for \$2.1 billion in total consideration. Dr. Kroeger was also previously the CEO of OvaScience, a publicly traded female fertility company where he led a turnaround effort resulting in a merger with Millendo Therapeutics. Earlier in his career, Dr. Kroeger led investing efforts as a Partner at The Aurora Funds, a venture capital firm focused on early-stage biotechnology and medical device companies, and held positions at Genzyme and Decision Resources.

Dr. Kroeger earned his bachelor's in Biology from Harvard University and his Doctor of Medicine from Stanford University School of Medicine. He did his Residency in General Surgery at the Brigham and Women's Hospital, an affiliate of Harvard Medical School, and earned his MBA from Harvard Business School. He currently serves on the Board of Directors of Stellaromics and Antlia Biosciences.



Erin Pennock Foff, MD, PhD
Chief Medical Officer

Dr. Foff is MapLight's Chief Medical Officer. A clinician with an extensive background in molecular biology and genetics, Dr. Foff brings with her more than 13 years' experience in conducting and designing clinical trials. She joins from Acadia Pharmaceuticals, where she served as executive director of clinical development and led the company's programs in the dementia/neurodegenerative space. Prior to joining industry, Dr. Foff held an academic physician-scientist position at the University of Virginia as a cognitive neurologist with a subspecialty interest in early-onset dementing illnesses and ran a translational laboratory program investigating diseases of RNA toxicity and neurodegeneration.

Dr. Foff holds a doctorate in molecular and human genetics from Baylor College of Medicine. She completed her medical degree at Jefferson Medical College, residency in neurology at University of Virginia, and holds a certification from the American Board of Psychiatry and Neurology.



James Lillie, PhD
Chief Scientific Officer

Dr. Lillie brings extensive cellular biology and biochemistry experience to MapLight with over 25 years of experience driving preclinical research and development across diverse therapeutic areas and indications. He joins MapLight from OvaScience where he served as the Chief Scientific Officer. Previously, Dr. Lillie served as Senior Director and, later, Vice President of In Vitro Biology at Sanofi Genzyme where he was responsible for developing a scientific strategy to create a sustainable, high value portfolio in rare diseases. He established a strong pipeline of small molecule drugs and supported the U.S. Food and Drug Administration's approval of Eliglustat and acceptance of two investigational new drug applications. Earlier in his career, Dr. Lillie held increasing roles of responsibility at Millennium Pharmaceuticals, Inc., was the Senior Director of Biology at Scriptgen Pharmaceuticals, and co-founded AMIRA, which was later sold to Repligen Corporation. He holds a doctorate from Harvard University, and a bachelor's in German Literature from Wesleyan University.

MPLT Management Team (2/2)



Anatol Kreitzer, PhD
Chief Discovery Officer

Dr. Kreitzer brings over 20 years of experience in academic neuroscience research, focused on mechanisms of neuromodulation and the application of optogenetics, electrophysiology, and imaging to understand neural circuit function and dysfunction. Prior to joining MapLight, Dr. Kreitzer was a Professor of Physiology and Neurology at the University of California, San Francisco, where he led a laboratory focused on the basal ganglia, a highly-conserved brain region in vertebrates that controls motivation, decision making, and the automation of behavior. His laboratory made important advances in the cellular, synaptic, and circuit mechanisms underlying motor deficits in Parkinson's disease and dyskinesia.

He holds a bachelor's degree from the University of California, Berkeley, and a doctorate in Neurobiology from Harvard University. He completed postdoctoral work at Stanford University, and advanced to Full Professor at UCSF, where he led the Neuroscience Graduate Program from 2016-2019.



Vishwas Setia
Chief Financial Officer

Vishwas Setia is MapLight's Chief Financial Officer. Prior to joining MapLight, Mr. Setia was a Managing Director in the healthcare investment banking group at Bank of America Securities. He brings a decade of investment banking experience advising biopharmaceutical companies and executing over \$15 billion in equity and equity-linked financings and \$100 billion in M&A and strategic transactions. He holds an M.B.A. from the Kellogg School of Management at Northwestern University and a B.E. in Mechanical Engineering from Delhi College of Engineering, India.



Jonathan Gillis, CPA, MSA
Chief Administrative and Accounting Officer

Mr. Gillis is MapLight's Chief Administrative and Accounting Officer. He brings over 15 years of combined public accounting and industry experience to the MapLight team. He joins from OvaScience, where he served as the Senior Vice President of Finance overseeing finance, human resources, information technology, and legal functions, and was instrumental in several public offerings and the ultimate merger with Millendo Therapeutics. Previously, Mr. Gillis was a Senior Manager of External Reporting and Compliance at GFI Software focusing on SEC compliance and preparing the company for its initial public offering. Earlier in his career, Mr. Gillis was a Manager at PricewaterhouseCoopers LLP and held responsibility for all stages of audit engagements with clients ranging from million-dollar private companies to multi-billion-dollar public companies.

He holds a B.S. and a M.S. in Accounting from Babson College.

MPLT Investment Thesis & Target Price Methodology/Risks

We rate MPLT shares Buy. MapLight Therapeutics is a clinical-stage biotechnology company focused on developing drugs for neuropsychiatric disorders. Our bullish thesis on MPLT is predicated on the potential of ML-007C-MA, a co-formulated drug consisting of the M1/M4 muscarinic agonist ML-007 and peripheral anticholinergic fesoterodine. MPLT's ongoing ph2 in schizophrenia is de-risked by KarXT's prior success leveraging a similar mechanism, though MPLT is also pursuing a ph2 in Alzheimer's disease psychosis where KarXT clinical data are still upcoming. Despite MPLT entering several years behind KarXT in an already crowded schizophrenia market, the "bar" for success is low and we've seen blockbuster sales for prior antipsychotics with minimal differentiation. In our view, less frequent dosing may be sufficient in itself for prescribers to select ML-007C-MA over KarXT, especially in ADP where KarXT's thrice daily dosing can be cumbersome. Ultimately though, this is CNS and good drugs with strong scientific rationale can still fail in clinical trials, but we think the risk/reward is upside-biased for this stock.

We derive a \$28 target price for MPLT shares based on our DCF analysis. Our DCF uses a 11% discount rate and 0% terminal growth rate, aligned with other similar stage companies in our coverage universe. In our model we risk-adjust MPLT revenues in SCZ and ADP at 50% and 20%, respectively, which reflects the level of scientific rationale and de-risking clinical data generated to date. We view both the SCZ and ADP as significant commercial opportunities if MPLT is successful. Our terminal growth rate of 0% takes into consideration the longer DCF forecast period extending up to 2038 which may be close to capturing peak sales of ML-007C-MA.

There are many risks to an investment in MapLight Therapeutics, including clinical, regulatory, and commercial risks, as well as the risk of a dilutive equity offering. Specifically, MPLT investors face the risk that ML-007C-MA and other pipeline programs could fail in clinical trials, be impacted by a safety setback, and/or be unable to achieve FDA approval. Beyond clinical trial and regulatory risks, MapLight, like other biotechnology companies, faces significant commercial risks due to competition, pressure from insurers, and/or a variety of other factors. Any of the above variables could impact the stock and our target price significantly.

Model Development Process and Alternative Approaches

Our model is built to replicate the company's reported financial information, with a particular focus on future drug product revenue and company profitability.

Data

SEC filings, company reports, FactSet data, medical literature and epidemiology, physician/key opinion leader feedback.

Model Theory

The model is built to forecast MapLight's income statement and cash flows. Our estimates are derived from forecasting future sales of ML-007C-MA in schizophrenia and Alzheimer's disease psychosis, with minimal, heavily risk-adjusted contributions from ML-004 in Autism Spectrum Disorder. We do not include any risk-adjusted revenues from other earlier-stage pipeline. We pair the revenue estimates along with our estimates of company profitability and capital requirements.

Model Processing Steps

Key user inputs in the model are indicated by a blue font color. Revenue, profit, and cash flow assumptions are replaced where necessary by realized data on a quarterly basis as the company makes these financials publicly available.

Qualitative Overlays and Assumptions

Forecasts for key assumptions are based in part on our expectations for clinical/regulatory success of MapLight's drugs, commercial demand and penetrance, and capital intensity. Our expectations stem from our assessments of preclinical and clinical data, conversations with physicians and researchers, published academic/medical literature and epidemiology, and our forecasts of future competitive dynamics within these therapeutic areas.

Model Limitations

Research recommendations, models, and estimates are of course limited in their accuracy of predicting future results. Model limitations are covered based on standard disclosures contained in each published research report.

MapLight Therapeutics (MPLT) - IS Data in \$MM, Except EPS	2023	2024	1H25	3Q25E	4Q25E	2025E	2026E	2027E	2028E	2029E	2030E	2031E	2032E
Product Revenue	-	-	-	-	-	-	-	-	-	-	-	15.5	81.0
Total revenue	-	-	-	-	-	-	-	-	-	-	-	15.5	81.0
COGS	-	-	-	-	-	-	-	-	-	-	-	1.2	6.5
R&D	49.7	68.5	46.6	25.0	26.0	97.6	110.0	100.0	120.0	130.0	140.0	90.0	90.0
SG&A	7.6	14.4	7.6	4.0	4.2	15.8	20.0	30.0	40.0	50.0	60.0	85.0	120.0
Other operating expenses	-	-	-	-	-	-	-	-	-	-	-	-	-
Operating expenses	57.3	82.9	54.2	29.0	30.2	113.4	130.0	130.0	160.0	180.0	200.0	176.2	216.5
Operating income	(57.3)	(82.9)	(54.2)	(29.0)	(30.2)	(113.4)	(130.0)	(130.0)	(160.0)	(180.0)	(200.0)	(160.7)	(135.4)
Interest income	1.1	4.5	1.5	1.0	1.0	3.5	-	-	-	-	-	-	-
Interest expense	-	-	-	-	-	-	-	-	-	-	-	-	-
Other income (expense)	0.0	1.8	0.5	-	-	-	-	-	-	-	-	-	-
Net interest and other income	1.6	5.4	2.0	1.0	1.0	3.5	-	-	-	-	-	-	-
EBIT	(55.7)	(81.1)	(52.2)	(29.0)	(30.2)	(113.4)	(130.0)	(130.0)	(160.0)	(180.0)	(200.0)	(160.7)	(135.4)
Net pre-tax income	(55.7)	(77.6)	(52.2)	(28.0)	(29.2)	(109.9)	(130.0)	(130.0)	(160.0)	(180.0)	(200.0)	(160.7)	(135.4)
Provision for income taxes	0.5	-	-	-	-	-	-	-	-	-	-	-	-
Net income (loss)	(56.2)	(77.6)	(52.2)	(28.0)	(29.2)	(109.9)	(130.0)	(130.0)	(160.0)	(180.0)	(200.0)	(160.7)	(135.4)
Diluted earnings per share (GAAP)	\$ (5.07)	\$ (6.27)	\$ (4.07)	\$ (2.14)	\$ (0.67)	\$ (5.34)	\$ (2.61)	\$ (2.55)	\$ (3.08)	\$ (3.40)	\$ (3.40)	\$ (2.68)	\$ (2.22)
Basic earnings per share (GAAP)	\$ (5.07)	\$ (6.27)	\$ (4.07)	\$ (2.14)	\$ (0.67)	\$ (5.34)	\$ (2.61)	\$ (2.55)	\$ (3.08)	\$ (3.40)	\$ (3.40)	\$ (2.68)	\$ (2.22)
Average shares outstanding, basic	11.1	12.4	12.8	13.1	43.6	20.6	49.9	50.9	51.9	52.9	58.9	59.9	60.9
Average shares outstanding, diluted	11.1	12.4	12.8	13.1	43.6	20.6	49.9	50.9	51.9	52.9	58.9	59.9	60.9

Source: Stifel estimates & company filings

MPLT DCF - SMM	2025E	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E	2038E	TV
CFO	(116)	(121)	(117)	(141)	(159)	(177)	(140)	(112)	7	135	297	399	484	532	
CFI	55	0	0	0	0	0	0	0	0	0	0	0	0	0	
Net Borrowing	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
FCF	(61)	(120)	(117)	(141)	(159)	(177)	(140)	(111)	8	135	297	399	484	532	4,835
Period	0.00	1.00	2.00	3.00	4.00	5.00	6.00	7.00	8.00	9.00	10.00	11.00	12.00	13.00	
PV FCF	(61)	(108)	(95)	(103)	(105)	(105)	(75)	(54)	3	53	105	127	138	137	1,245

Total FCF, PV	1,102.5
Net Cash 4Q25E	462.0
Implied Market Capitalization	1,564.5
Diluted Shares Outstanding 4Q25E	55.9
NPV Per Share	\$28

Discount Rate	11.0%
Terminal Growth Rate	0.0%

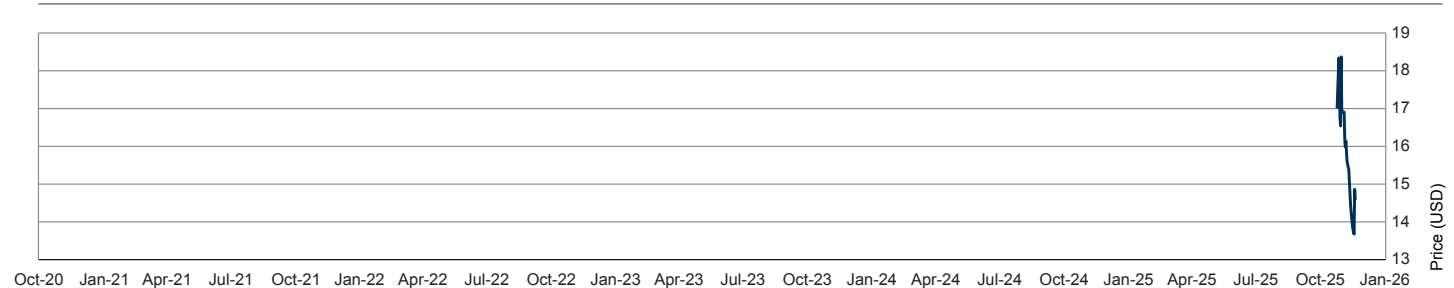
Source: Company Filings, Stifel estimates. Note: Diluted share count includes options and RSUs.

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MapLight Therapeutics, Inc. (MPLT) as of November 19, 2025 (in USD)



*Represents the value(s) that changed.

Buy=B; Hold=H; Sell=S; Discontinued=D; Suspended=SU; Discontinued=D; Initiation=I

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